

EXHIBIT J – CONTRACTOR SITE ENVIRONMENT, HEALTH AND SAFETY REQUIREMENTS

These requirements apply to all Contractor and Subcontractor personnel working under this Agreement, including casual or temporary employees.

PRIOR TO COMMENCEMENT OF WORK

Contractor shall provide Siemens Energy with a copy of Contractor's company environmental, health and safety policies and procedures (hereinafter "Contractor EHS Program"), a site-specific environmental, health and safety plan, and other EHS deliverables in accordance with the timeframes stated in the "Siemens Energy Contractor EHS Requirements" attached hereto as Attachment A. These will be reviewed by Siemens Energy for consistency with applicable industry standards and Siemens Energy requirements. In all cases, the most stringent of the requirements shall be applied to the Work. Contractor shall perform all work in a safe and environmentally sound manner, in compliance with Applicable Laws and the environment, health and safety requirements of this Agreement.

Contractor will attend a Siemens Energy pre-start environmental, health, and safety review meeting prior to mobilization. This meeting is the opportunity for Contractor to present, review, and discuss applicable environmental, health and safety requirements for the Project. Attendance for this meeting is mandatory for all Key Personnel as identified by Siemens Energy.

As one of Contractor's Key Personnel, Contractor shall appoint a dedicated environmental, health, and safety professional to manage the Contractor EHS Program and additional EHS staff resources in accordance with the requirements of this exhibit. In the event Siemens determines that additional dedicated environmental, health and safety professionals are needed in the Contractor organization, Contractor agrees to provide such personnel.

EHS WORKSHOPS

Siemens Energy expects Contractor's full commitment in maintaining a safe and healthful project. Failure to comply with operational or administrative environmental, health, and safety requirements may result in an immediate order to stop Work and/or payment until the issues are resolved, at the Contractor's expense.

Prior to proceeding with any Work on the Project and at specified intervals during an ongoing project, it is a requirement that Contractor and Subcontractors actively participate in EHS workshops, including a pre-mobilization Safe-Start Workshop, with Siemens Energy. Contractor shall conduct additional Safe Start Workshops with each of their Subcontractors, with participation and input from Siemens Energy.

This meeting will be used to outline Siemens Energy and Project specific expectations for your organization, as it relates to environmental, health, and safety. In addition, it shall provide the Contractor a forum to identify major phases of Work and the control measures Contractor will employ to control the hazards associated with the Work.

ATTACHMENT A
SIEMENS ENERGY CONTRACTOR EHS REQUIREMENTS

1. INTRODUCTION, PURPOSE.....	69
2. SCOPE OF APPLICABILITY, VALIDITY	69
3. ABBREVIATIONS, DEFINITIONS	69
4. ACCIDENT REPORTING AND INVESTIGATION	70
5. CONTRACTOR EHS MANAGEMENT	71
6. CONTRACTOR EHS STAFF	76
7. CONTRACTOR'S SITE SPECIFIC EHS PLAN	77
8. CORRECTIVE ACTIONS / STOP WORK	78
9. EHS MANAGEMENT AUDIT PROCESS	78
10. EHS PERFORMANCE	79
11. EMERGENCY RESPONSE PLAN	79
12. INJURY AND CASE MANAGEMENT	80
13. INSPECTION PROGRAMS	80
14. SHORT SERVICE EMPLOYEES.....	81
15. SITE ACCESS AND SECURITY.....	81
16. ZERO HARM WORKSHOPS	82
17. ZERO TOLERANCE POLICY	82
SAFETY:	83
18. ABRASIVE BLASTING / SPRAY COATING	83
19. BUCKET TRUCKS AND DIGGER DERRICKS	84
20. CHAIN SAWS	84
21. COMPRESSED GAS CYLINDERS.....	84
22. CONFINED SPACE ENTRY	86
23. CRANES AND RIGGING	90
24. CUTTING, GRINDING AND WELDING OPERATIONS	96
25. DIVING AND WATER OPERATIONS.....	98
26. DROPPED OBJECT PREVENTION PROGRAM	99
27. ELECTRICAL SAFETY AND ENERGY CONTROL	101
28. ELEVATING WORK PLATFORMS (EWP)	105
29. EQUIPMENT, VEHICLE, AND TOOL INSPECTION	107
30. EXCAVATIONS AND GROUND PENETRATIONS	107
31. EARTHWORK OPERATIONS	110
32. HAUL TRUCKS / DELIVERY TRUCKS	111
33. FALL PROTECTION	113
34. FIRE PREVENTION & PROTECTION.....	113
35. GAS BLOW	116
36. HAND AND POWER TOOLS.....	116

Subject to Management Review and Approval

37. JOB-MADE STANDS	117
38. LADDERS AND SCAFFOLDS..... Fall Protection required if climbing a Ladder > 12 ft.	118
39. MACHINE GUARDING	121
40. MATERIAL HANDLING	121
41. PERSONAL PROTECTIVE EQUIPMENT..... Boots must be Leather ; Spoggle required under shields	122
42. POWERED INDUSTRIAL TRUCKS (FORKLIFTS)	124
43. SIGNS AND BARRICADES..... All Barricades must be Tagged - "Danger and Caution"	126
44. SITE INSTALLATIONS.....	127
45. STEEL ERECTION..... No Walking Steel - Unless Engineered Life Line is Used	128
46. STORM AND LIGHTNING	128
47. VEHICLES AND EQUIPMENT	129
48. WORK ZONE SAFETY / PUBLIC ROADS	131
OCCUPATIONAL HEALTH	132
49. ASBESTOS CONTAINING MATERIALS.....	132
50. AVOIDING INSECTS AND SNAKES.....	132
51. BLOOD-BORNE PATHOGENS	132
52. ELECTROMAGNETIC FIELDS (EMF) AND MEDICAL DEVICES.....	132
53. EYEWASH AND EMERGENCY SHOWERS.....	134
54. HAZARD COMMUNICATION.....	134
55. HEARING CONSERVATION	134
56. HEAT AND COLD EXPOSURE	136
57. HEXAVALENT CHROMIUM	138
58. LEAD IN CONSTRUCTION	138
59. RADIOGRAPHY	138
60. RESPIRATORY PROTECTION.....	139
61. SUBSTANCE ABUSE PREVENTION	140
62. WELDING AND THERMAL CUTTING – HEALTH EXPOSURES.....	141
ENVIRONMENTAL PROTECTION	142
63. ENVIRONMENTAL PLANS	142
64. SPILL PREVENTION	142
65. CLEAN UP AND DISPOSAL	144
ADMINISTRATIVE DOCUMENTS	145
66. CROSS REFERENCES.....	145
67. STANDARDS	145

1. Introduction, Purpose

- 1.1. The requirements contained in this document ("Requirements") are applicable to all Siemens Energy contractors, all tiered subcontractors, and consortium partners, their contractors and all their tiered subcontractors. If any exception is required in regard to the applicability of certain sections or requirements, such as for limited scope of work, these shall be determined by a joint collaboration between Siemens Energy, Contractor and, if applicable, consortium partners, the final decision shall be made by Siemens Energy and consortium partner.
- 1.2. These Requirements are in addition to those contained in the OSHA Standards (e.g., 29 CFR 1926 and 1910) and or U.S. Army Corps of Engineers' Safety and Health Requirements Manual EM 385-1-1. This last one will be applicable for all construction projects with the Department of Defense. These requirements are also supplementary to the requirements contained in the Agreement, technical contractual conditions and Siemens Energy safety procedures as part of continuous improvement efforts to provide a healthy and injury-free workplace to our employees, Contractors, and the communities we serve.
- 1.3. If a conflict exists between the Requirements and the Contractor requirements or Siemens Energy requirements, then a variance can be permitted by a joint collaboration of Siemens Energy, consortium partner, and Contractor with the final decision made by Siemens Energy and the Consortium partner.

2. Scope of Applicability, Validity

- 2.1. The Requirements, which are applicable to the Contractor and all Subcontractors, shall be addressed and included within the Contractor project site specific EHS plan(s).
- 2.2. The Requirements do not apply to delivery services vendors such as FedEx, UPS, compress gas deliveries, vending machine personnel, etc. However, vendors will be required to adhere to basic safety requirements as developed in the site-specific plan.
- 2.3. The Requirements are valid as of the date of issue indicated on the title page. Unless otherwise required, The Requirements shall be reviewed at intervals, not to exceed two (2) years from the date of issue. It shall be revised if enhancements become available or if changes in the business conditions dictate. Contractors and Subcontractors will adhere and comply with the most current official version of this document at no additional cost to Siemens Energy.

3. Abbreviations, Definitions

EHS Environmental, Health, and Safety

Qualified	Having possession of a recognized degree, certificate, or professional standing, or by extensive knowledge, training and experience, has successfully demonstrated the ability to solve or resolve problems relating to the subject matter, the Work, or the Project.
Competent Person	One who is capable of identifying existing and predictable hazards in the surroundings or working conditions which are unsanitary, hazardous, or dangerous to employees and who has authorization to take prompt corrective measures to eliminate them
High Risk Activities	Activities that are of great risk including falls from height, lifting operations, being trapped, flying or falling objects, machinery use, electrical contact, drowning or asphyxiation, contact with harmful substances, confined spaces, simultaneous operations, excavations, hot work and other activities as determined by Siemens Energy.

4. Accident Reporting and Investigation

- 4.1. All incidents (e.g., injuries, illnesses, near miss, property damage, and vehicle) and environmental events (e.g., chemical / petroleum release, leak, or spill) occurring on the Project shall be reported, investigated, and analyzed by a Competent or a Qualified person. All injuries, illnesses, and near misses shall receive a causal analysis with contributing factors identified. All injuries and illnesses shall be reported regardless of severity. Even if no onsite first aid treatment or offsite medical care is provided, Contractor shall report to Siemens Energy any injury or illness.
- 4.2. Contractor shall maintain a log of all injuries and illnesses. This data shall be used by the Contractor to conduct trending analysis and research to prevent the recurrence on such incidents.
- 4.3. Recordable injuries, lost time accidents, serious injuries and fatalities, and environmental events reported to a government agency shall receive a root cause analysis. In some cases, Siemens Energy may require the Contractor to perform a root cause analysis depending on the significance of the near miss or first aid case. Contractor shall develop corrective actions to prevent recurrence of environmental events, first aid injuries, near misses, recordable injuries, and lost time accidents and monitor the effectiveness of corrective actions.
- 4.4. Contractor shall implement a formal root cause, causal analysis, and corrective action process that has been submitted to Siemens Energy for acceptance or attend Siemens Energy's site training and follow Siemens Energy's guide for root cause analysis.
- 4.5. Near misses, first aids, recordable injuries, and lost time accidents are to be subject to trending and common cause analysis.
- 4.6. Where Siemens Energy is the lead consortium partner, in the event of a serious incident, the incident scene shall not be disturbed until released by Siemens, except for circumstances where "imminent danger" exists to those performing any emergency services.
- 4.7. Contractor shall immediately notify Siemens Energy within one (1) hour of any incident, injury, or environmental event. Contractor will immediately notify Siemens Energy Project Director / Manager, Site Manager, and onsite Siemens Energy Lead EHS Professional by cell phone, email, or direct contact (e.g., face-to-face).

- 4.8. Before shift's end, Contractor shall provide a written one-page summary of the incident including the date and time of the event, name of affected worker(s), event description, and available details.
- 4.9. A detailed written report including photographs, identified causes, and recommended corrective actions shall be submitted to Siemens Energy within 24 hours unless otherwise agreed to by Siemens Energy.
- 4.10. Contractor shall ensure all Subcontractors are informed and comply with the Siemens Energy incident reporting and post-incident case management requirements.
- 4.11. Siemens Energy reserves the right to observe or participate in any environmental, health, and safety accident investigation conducted by Contractor or anyone performing work for or on behalf of Contractor.
- 4.12. Siemens Energy reserves the right, at its sole discretion, to initiate its own accident investigation.
- 4.13. Contractor shall immediately correct any unsafe or environmentally hazardous conditions created by Contractor. In the event Contractor fails to immediately correct such unsafe or environmentally hazardous conditions, Siemens Energy may either have the unsafe or environmentally hazardous conditions corrected by others at Contractor's expense, or direct the Work be stopped in the area of the unsafe condition. This right to stop the Work shall not give rise to any duty to Contractor or to any third party on the part of Siemens Energy to exercise this right. Any Work stoppage for unsafe conditions or environmentally hazardous conditions is at no cost to the Siemens Energy.
- 4.14. Contractor shall create an investigation report after analyzing the incident root causes for all MTC / Recordable Injuries within 72 hours. Other incidents may also require a root cause investigation.
- 4.15. The report must ONLY contain facts as they are known at the time of the report.
- 4.16. Corrective actions (how to prevent recurrence) must be included in the final report.
- 4.17. The corrective actions must include the site management representative responsible for completing the action, as well as an expected completion date. All corrective actions identified must be assigned to a person responsible to ensure completion, as well as a completion due date
- 4.18. Contractor shall maintain a list of incidents and their corrective and preventive actions with date and proof of implementation.

5. Contractor EHS Management

- 5.1. Contractor shall review all Subcontractor and sub-tier Contractor site specific environmental, health, and safety programs and risk assessment and method statements (RAMS) for compliance with the Project EHS requirements established by Exhibit J – Contractor Project Environmental, Health and Safety Requirements. Contractor shall provide all Subcontractors a copy of all Project environmental, health, and safety-related requirements and establish written verification of such, subject to Siemens Energy review. Contractor shall ensure all Subcontractors are Qualified per the Siemens Energy Contractor Qualification procedure / program. Siemens Energy has the right to reject any and all Subcontractors to the extent applicable under the

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Agreement. Contractor shall furnish the following information to Siemens Energy for each Subcontractor:

- 5.1.1. Experience Modification Rating (EMR) average for the last 3 years
- 5.1.2. Workplace fatalities within the last 3 years
- 5.1.3. Detailed explanation of all OSHA, environmental agency, or any other governmental citations related to EHS received in the last 3 years
- 5.1.4. Company's OSHA Total Recordable Incident Rate (TRIR) for the last 3 years
- 5.1.5. Company's Day Away Restricted Time (DART) rate for the last 3 years
- 5.1.6. Company's Lost Time Incident Rate (LTIR) for the last 3 years
- 5.2. Contractor Site EHS Orientation
 - 5.2.1. Contractor's orientation and Siemens Energy site orientation program shall be provided to all Contractor employees, Subcontractors, and vendors on site. This orientation shall be documented and records kept by Contractor, subject to review by Siemens Energy. Any individual performing Work on the Project, regardless of anticipated time on site, shall receive the Contractor and Siemens Energy's orientation.
 - 5.2.2. The Contractor orientation shall be submitted to Siemens Energy for review and acceptance at least 30 days prior to mobilization.
 - 5.2.3. Contractor's orientation program may make provisions for a Visitor's Orientation for individuals who will be escorted at all times and on site for no longer than 5 days. A visitor is defined as a person meeting the above criteria and not performing any physical labor tasks, i.e., management inspections, audits, etc. Visitor orientation program shall be submitted to Siemens Energy for review and acceptance and must include a brochure style handout highlighting Project specific EHS requirements pertinent to the visitor. Each visitor orientation must include a means of verification of understanding, which must be made available to Siemens Energy upon request.
- 5.3. Contractor and Subcontractor onsite first line supervision (e.g., foreman/supervisor level) shall have successfully completed OSHA 30-Hour Construction Outreach training or, if outside the U.S., equivalent nationally recognized training.
- 5.4. Contractor shall develop and implement a supervisor training program for all supervision. This training must be completed prior to commencement of supervisor duties and shall include, at minimum, the following:
 - 5.4.1. Supervisor Responsibilities for Safety Management
 - 5.4.2. Safe Work Practices
 - 5.4.3. Toolbox Meetings
 - 5.4.4. Emergency Procedures
 - 5.4.5. First Aid Procedures, First Aid, CPR, & AED
 - 5.4.6. Incident Reporting and Investigation
 - 5.4.7. Fire Protection and Prevention
 - 5.4.8. New Worker Orientation
 - 5.4.9. Permit System & LOTO
 - 5.4.10. Confined Space
 - 5.4.11. Crane and Rigging Procedures

- 5.4.12. Risk Assessment and Method Statement
- 5.4.13. Field Level Hazard Analysis (FLHA)
- 5.4.14. Safety Observations/Inspections
- 5.5. Contractor and Subcontractor onsite management (superintendents & managers) shall have successfully completed OSHA 30-Hour Construction Outreach training or, if outside the U.S., equivalent nationally recognized training, unless otherwise accepted by Siemens Energy.
- 5.6. Contractor shall maintain a training matrix outlining required training for Contractor and Subcontractor personnel on site. This training matrix shall list each person's name and completion of pertinent training. Contractor's database, spreadsheet, or form shall be submitted and approved by Siemens Energy. An example training matrix can be provided by Siemens Energy, if necessary.
- 5.7. Contractor shall submit to Siemens Energy sufficient documentation to show all personnel on the site have received appropriate EHS training and that such training is current. Such documentation shall include, but not limited to, names of personnel, specific EHS training course titles, and date each person completed the specific course. Siemens Energy reserves the right to conduct an audit of training course content, records, and delivery methods, as necessary.
 - 5.7.1. Training matrix shall include the completion dates of all required training (e.g., PPE, hazard communication, control of hazardous energy / lock out tag out, confined space entry, fall protection, scaffold user, crane operator, signal person, rigger, hearing conservation, respiratory protection, powered industrial truck, orientation, fire watch, spotter, and dropped object prevention etc.).
 - 5.7.2. Additional onsite training (e.g.,) shall also be provided to personnel depending on their job duties and assigned tasks. This additional training shall be included in the training matrix with documentation provided to Siemens Energy upon request for review.
 - 5.7.3. Training dates and certifications shall be provided for mobile equipment operators (e.g., earthmoving equipment, aerial lift, skid steer loader, and powered industrial truck operators).
- 5.8. Contractor shall maintain and submit to Siemens Energy a list of Competent Persons (e.g., scaffolding, trench and excavation, confined space entry, electrical safety, fall protection, cranes and rigging, etc.) accompanied with a relevant work history/ experience and training that qualifies them to perform the duties of a Competent Person and/or a certificate of completion from a recognized training provider or employer audited program, as agreed upon with Siemens Energy.
- 5.9. Contractor shall provide a weekly environmental, health, and safety performance report by 12:00 PM (EST) on Monday of the following week. The report shall summarize the following information on a weekly monthly, year-to-date, and Project-to-date basis.
 - 5.9.1. Number of personnel on site at the end of the week
 - 5.9.2. Hours worked on site (including all Subcontractors)
 - 5.9.3. First aid incidents
 - 5.9.4. Doctor visit cases (i.e., incidents requiring offsite medical evaluation or treatment)
 - 5.9.5. Recordable injuries
 - 5.9.6. Lost time / lost workday and restricted duty cases

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- 5.9.7. OSHA incident rates
 - 5.9.8. Vehicle and equipment accidents by type
 - 5.9.9. Near miss incidents
 - 5.9.10. Fire Events
 - 5.9.11. Environmental events (e.g., spills / leaks, reportable environmental incidents, and negative audit findings or notice of violation by a regulatory agency/ governing body).
 - 5.9.12. Trending analysis for all incidents and safety observations
- 5.10. The monthly report shall also include a narrative summary of recordable incidents, first aid cases, near misses, equipment / vehicle incidents, summary of industrial hygiene monitoring, and non-occupational events.
- 5.11. The monthly report shall also include a summary of safety observation data and information, corrective actions, and/or improvement actions Contractor is taking or planning to implement to ensure the effectiveness of the EHS management program.
- 5.12. Risk Assessments are an integral part of the injury prevention system. Contractor shall develop a Risk Assessment program consisting of two phases.
- 5.12.1. Phase 1- The Risk Assessment and Method Statement outlines the specific processes and safety procedures to be followed when carrying out high-risk activities. RAMS must be prepared by supervision well in advance of the scheduled work.
 - 5.12.2. Phase 2 - the Field Level Hazard Analysis is a task specific process of risk assessment and mitigation performed by the work crew at the time the actual task will be performed.
- 5.13. Contractor and Subcontractors shall develop a RAMS for High-Risk Activities such as falls from height, lifting operations, being trapped, flying or falling objects, machinery use, electrical contact, drowning or asphyxiation, contact with harmful substances, confined spaces, simultaneous operations, excavations, and hot work. Contractors shall provide completed RAMS to Siemens Energy 10 days prior to planned Work activities and job operations for review and acceptance.
- 5.14. Contractor must provide workers with any RAMS related to their assigned task so that it can be used during the completion of their FLHA. Both the RAMS and the FLHA must be posted in the immediate vicinity of the task being performed.
- 5.15. A FLHA, or equivalent, shall be completed by assigned workers immediately prior to performing the task. Specific task steps, hazards, controls, and any other pertinent information shall be included in the FLHA.
- 5.16. Contractor shall implement a Safety Observation Program, which includes behavioral based safety elements, that is acceptable to Siemens Energy.
- 5.16.1. Contractor shall ensure that management personnel and all supervision are trained in the program elements and perform a minimum of one safety observation per day.
 - 5.16.2. Contractor shall designate 5 percent of their craft workers to be trained in conducting peer to peer safety observations. Each of these trained craft workers shall conduct a minimum of one safety observation per day for 2 weeks, after which a new set of craft workers will be designated and trained to perform observations. This shall continue through the end of the project.

What does this mean?

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5.16.3. Contractor shall develop and institute a process for tracking and trending observation findings on a weekly basis. Trending data shall be used to develop safety training schedules, meeting agendas, campaigns, etc.

5.16.4. If Contractor does not have an adequate BBS / Safety Observation Program, the Siemens Energy's' observation program may be utilized by Contractor, as mutually agreed by Contractor and Siemens Energy.

5.17. Contractor shall implement a site Health and Safety Committee that is comprised of trade members from each trade and Subcontractor on the site. The committee ratio shall be a minimum of 1:50 workers per craft (e.g., laborers, carpenters, iron workers, pipe fitters, electricians, and etc.) and each Subcontractor will have at least one representative attend the committee meetings, which shall be conducted every two weeks, or as local (region/country) requirements dictate.

5.18. Contractor shall implement a daily stretching program (i.e., Stretch and Flex) as part of the daily pre-task meeting process.

5.19. Contractor shall conduct weekly joint labor / management safety walk down inspections of the worksite. The inspection team shall meet at a regularly scheduled date and time and be comprised of craft / hourly workers, supervisors, and EHS representatives. Findings and corrective actions from these inspections shall be documented and made available to Siemens Energy upon request.

5.20. Contractor shall conduct weekly management safety walks for the purpose of identifying and mitigated hazards and promoting a positive safety culture through direct engagement with workers.

5.21. Contractor shall conduct daily health and safety inspections of the Work and Subcontractors' work and implement immediate corrective actions for unsafe conditions or unsafe behaviors. Contractor shall document inspections of the worksite and make available to Siemens Energy upon request.

5.22. A single, comprehensive Action Item list or register shall be maintained for safety findings at the Project by the Contractor to document findings or deficiencies not immediately corrected during inspections to ensure open items are monitored and tracked through to completion. This list or register shall be available to Siemens Energy upon request for effectiveness review to ensure hazards are being corrected and items are completed. The list shall be updated weekly (minimum). Open issues which cannot be immediately closed and can cause injury to people or property must be temporarily mitigated upon discovery. Open issues must be closed within 10 days unless otherwise agreed upon by Siemens Energy.

5.23. Contractor shall conduct a weekly project-wide safety meeting for all onsite workers including Subcontractors. This meeting shall review recent issues, incidents, job observation trends, upcoming work, and relevant EHS topics. The weekly safety meeting is conducted in addition to the daily pre-job briefing. Topics reviewed and attendees must be documented.

5.24. Contractor shall facilitate EHS meetings every two weeks with Contractor's EHS personnel and Contractor's foremen to ensure effective communication exists between EHS and front-line supervision.

← Incorporate with Weekly Coordination Meeting

5.25. Special circumstance meetings (e.g., safety stand downs) shall be conducted, at no cost to Siemens Energy, by the Contractor as needed (i.e., may be requested by Siemens Energy). These meetings will be conducted when an accident(s), near miss

Weekly All-Hands

incident(s), recurring safety trends, or industry event occurs that warrants a special safety meeting and communication with onsite workers.

- 5.26. For primarily English language countries, if Contractor or its Subcontractor(s) employs personnel who are not fluent in speaking, reading, or understanding English, Contractor or Subcontractor shall provide (onsite at all times) a minimum of one bilingual worker with the ability to communicate in English for safety purposes for every five (5) non-English-speaking workers at the site. Siemens Energy may require additional bilingual personnel by Contractor for safety purposes depending on the duties assigned to personnel or the hazards at that site.
- 5.27. For countries whose primary language is not English, contractor shall have sufficient numbers of personnel fluent in English to act as translators.
- 5.28. Bilingual workers (i.e., translators) shall be identified by a unique identifier (e.g., stripes on hard hat, different color hard hat, different color vest, sticker, etc.).
- 5.29. Training shall be translated / provided in the appropriate language, which non-English speaking workers can comprehend and understand.
- 5.30. Contractor shall immediately notify Siemens Energy of any agency inspection (e.g., federal, state, or local) as soon as they become aware the agency inspector / compliance officer is on the site.
- 5.31. During commissioning stage and operation at site, Contractor and Subcontractors shall follow Siemens Energy site specific policies and requirements regarding jewelry, such as necklaces, rings, earrings, bracelets, and watches. Medical alert identification/bracelets are exempt. Jewelry shall not be worn where contact with energized lines or equipment is possible.

6. Contractor EHS Staff

- 6.1. If Contractor has less than 25 employees on site, they may assign EHS duties to a Foreman/Supervisor in addition to their regular assignments.
- 6.2. A full-time on site EHS professional is required when a project exceeds 25 employees. If project conditions, activities or management structure indicate the need, Siemens Energy reserves the right to require a full-time safety person at a lower work force level.
- 6.3. When a project exceeds 50 employees, a second safety professional is required. Each additional 100 employees require the addition of another safety person.
- 6.4. Qualified full-time EHS professionals shall have no duty other than to ensure full compliance with the Siemens Energy-accepted, Contractor site specific EHS plan for the prevention of incidents and injuries and to address unsafe and undesirable behavior or conditions.
- 6.5. Siemens Energy reserves the right to review and approve the qualifications of all EHS professionals prior to assignment on the project.
- 6.6. Unless otherwise agreed to by Siemens Energy EHS Department (Americas), EHS professionals assigned to the project shall be qualified as follows:
 - 6.6.1. EHS Manager (lead) – baccalaureate level degree in a safety-related field or relevant experience, professional certification / designation (i.e., CSP, ASP, CHST, local/nationally recognized certification); and 5 plus years of full time

EHS experience in large-scale industrial construction. Additionally, the EHS manager must have training in incident investigation / root cause analysis, first aid, CPR, and AED.

- 6.6.2. EHS Specialist (Midlevel) – OSHA 510 for Construction or equivalent nationally recognized training; 2–3-years of experience in large-scale industrial construction; and first aid, CPR, and AED training.
- 6.6.3. EHS Technician (Entry Level) – OSHA 30-Hour Construction Outreach training; First aid, CPR, and AED training. Note: It is Siemens Energy's expectations that entry level EHS technicians are only assigned entry level duties and cannot be tasked with EHS oversight of complex or High-Risk Activities.

7. Contractor's Site Specific EHS Plan

- 7.1. Contractor and Subcontractors shall develop and submit a project-specific Environmental Health and Safety Plan to Siemens Energy for review 60 days prior to mobilization. At a minimum, all sections outlined in this document applicable to the Contractor's scope of Work and the fundamental principles addressed in this document shall be included in the Contractor's project-specific EHS plan. The plan should include the following:
 - 1. Contractor's general company information with a corporate and Project org-chart, including contact details.
 - 2. Contractor sanitary and Site facility provisions for its personnel.
 - 3. Description of the Contractor's scope of works. (If Contractor has a large scope of Work, such as the main civil or erection works, the EHS plan may be revised as the phases of their Work develops).
 - 4. A detailed risk analysis of the Work to be performed.
 - 5. The plan should include health hazards such as, but not limited to the following:
 - a) Construction health and safety hazards
 - b) Noise and vibration hazards
 - c) Biological hazards; those which would cause unsanitary conditions
 - d) Chemical hazards
 - e) Respiratory hazards, which include hazardous dusts, mists, vapors, gases, fumes, smoke, or any hazardous particulate.
 - 6. Identification listing of radio frequencies to be used and Siemens Energy acceptance to ensure no interference with plant equipment or Siemens Energy radio frequencies at the site.
- 7.2. In addition to the Contractor's site specific EHS plan, more detailed standalone plans/ programs, such as Crane and Rigging Management, Steel Erection, Fall Protection, Confined Space Entry, and Control of Hazards Energy / Lockout Tagout (LOTO) may be submitted separately to address more complex work practices and/or procedures. However, the fundamental requirements (i.e., principal guidelines) of these plans shall be included and addressed in the Contractor's site specific EHS plan.

Note: Contractor shall be responsible to ensure that its personnel are fully informed of Contractor's EHS plan content and of their individual responsibility to respect the work practices and safety measures described therein.

8. Corrective Actions / Stop Work

- 8.1. Contractor must have an employee Stop Work Policy which clearly authorizes every worker on site to stop any job which has the potential to cause injury or damage without fear of reprisal. Contractor must ensure that all employees are aware of this policy through training, posters, etc.
- 8.2. Siemens Energy reserves the right to require the Contractor to address unsafe, environmentally hazardous, and unhealthy working conditions including taking corrective action when unsafe and unhealthy working conditions are observed (e.g.: lack of good housekeeping practices, use of equipment in poor conditions, failure to adhere to Siemens Energy regulations, inadequate monitoring, etc.) Contractor shall take action any time there is a risk of harm to personnel, the environment or material resources and/or where any non-compliance with any Applicable Laws or any Project related regulations or instructions exists.
- 8.3. Siemens Energy has the right to direct the Contractor to remove from the worksite any person, property, or equipment that, in Siemens Energy's opinion, is deemed unsafe.
- 8.4. The stop work order shall remain in force until such time that corrective measures have been implemented.
- 8.5. Corrective measures may include, but not be limited to, putting collective or individual protection into place, cleaning-up (housekeeping) and re-organization of the work area, retraining of staff, removal or replacement of managers and supervisors and/or method statement and risk assessment review.
- 8.6. Taking into consideration the severity of the circumstances a stop work order may be issued by the Siemens Energy Site Management SSM as follows:
 - 8.6.1. Verbal stop work order: SSM will inform the Contractors supervisor directly in the work area/s in question to suspend all or some of the work activities until corrective measures have been implemented.
 - 8.6.2. Formal stop work order: The SSM will inform the Contractors supervisor directly in the work area/s in question to cease all work activities. The Contractor's Site management will be notified in writing by the SSM, and a copy shall be transmitted to the Contractor's Project management. The Contractor's work shall not be permitted to re-start until the SSM is satisfied that permanent corrective measures have been implemented.
- 8.7. Work stoppage is at no cost to Siemens Energy.

9. EHS Management Audit Process

- 9.1. Contractor shall conduct comprehensive audits of its site EHS programs to ensure compliance with the requirements set forth in this document and all applicable Agreement requirements. These audits shall be conducted by Contractor corporate office management every 6 months unless otherwise agreed to by Siemens Energy.
- 9.2. Siemens reserves the right to require additional audits at no cost to Siemens Energy. Contractor shall initiate the additional audits within the time specified by Siemens Energy at the time of the request.

10.

EHS Performance

- 10.1. In the event Contractor or a Subcontractor experience ongoing environmental, health, and safety concerns as determined by Siemens Energy, Contractor shall be subject to the following action at Siemens Energy discretion:
 - 10.1.1. Submitting a written Environmental, Health, & Safety Recovery Plan to Siemens Energy detailing what improvements will be made to their EHS program and a timeline as to when the changes will be implemented.
 - 10.1.2. Siemens Energy may require Contractor, at no additional cost to Siemens Energy, to increase its number of EHS professionals until the Project stabilizes to a safe condition as determined by Siemens Energy.
 - 10.1.3. Work Stoppage is at no cost to Siemens Energy.
 - 10.1.4. Additional training or other resources, at no cost to Siemens Energy.
 - 10.1.5. To the extent applicable, termination per the Agreement to which these requirements apply.

11.

Emergency Response Plan

- 11.1. Contractor shall provide an Emergency Response Plan or Emergency Action Plan and shall communicate this plan to all their site personnel. Contractor shall submit this plan to Siemens Energy or its designee for review, comment, and acceptance 14 days before mobilization and provide updates as available. This plan shall include the following as a minimum:
 - 11.1.1. Identification of the possible emergencies to face during the execution of the project and related to the scope of work.
 - 11.1.2. Reporting emergencies including medical emergencies, fire emergencies, chemical spills / releases, rescue from heights (e.g., fall rescue plan), confined space rescue, bomb threats, severe weather, and etc.
 - 11.1.3. Assembly points, evacuation routes, and accounting for personnel procedure (i.e., ensuring everyone is accounted for in an evacuation or emergency) shall be included in the plan.
 - 11.1.4. Telephone numbers and locations of facilities including, but not limited to, hospitals, physicians, police, fire department, and emergency medical services shall be provided.
 - 11.1.5. Locations of work areas shall be addressed in the Contractor site-specific Emergency Response Plan (bilingual as necessary). Contractor is responsible for providing adequate safe zone facilities / shelter for their employees, Subcontractors and Siemens Energy personnel. Contractor shall liaise with Siemens Energy in identifying assembly points.
 - 11.1.6. Drill program for different events.
 - 11.1.7. Contractor's Emergency Response Plan shall include emergency equipment available on site to respond to an emergency.
 - 11.1.8. A Layout of the area including muster points, evacuation routes and emergency equipment shall be attached to the plan.

12.

Management

Injury and Case

- 12.1. Except for medical emergencies where an ambulance is required, Contractor or Subcontractors shall transport work-related injured or ill employees to and from medical facilities initially and remain with the injured or ill person throughout this period.
- 12.2. If offsite medical treatment is required, Contractor shall appoint appropriate personnel to accompany the injured or ill employee to the offsite medical provider.
- 12.3. Individuals who transport injured or ill workers off site for treatment shall have First Aid / Cardiopulmonary Resuscitation (CPR) training and general knowledge of the OSHA Record keeping standard regarding work restrictions and the option employees have to ask attending medical professionals if over-the-counter medication maybe taken in lieu of prescription medication.
- 12.4. Subcontractors shall notify Contractor before taking any worker off site for medical care or evaluation.
- 12.5. Contractor shall notify Siemens Energy prior to sending anyone off site for medical treatment or examination.
- 12.6. At least one person shall be available at the worksite at all times to render First Aid/ CPR. This person shall have a valid certificate in First Aid, CPR, and Automated External Defibrillator (AED) training. A minimum ratio of one such Qualified person for every 50 workers shall be maintained throughout the Project.
- 12.7. Contractor shall ensure an adequately supplied medical jump bag (i.e., EMT / trauma bag) and AED are provided and available to First Aid / CPR / AED trained personnel.
- 12.8. Contractor shall ensure First Aid / CPR / AED trained personnel are capable of responding promptly (i.e., 3-4 minutes) in areas where serious, life-threatening injuries or illnesses (e.g., suffocation, severe bleeding, falls, electrocution, or amputation) are possible and/or reasonably anticipated to avoid permanent medical impairment or death.
- 12.9. Remove Section Entirely.

13.

Inspection Programs

- 13.1. Contractor shall develop a project specific inspection program using green, yellow, red and blue color tape over the course of each year as it follows:
 - 13.1.1. Green tape from January to March, yellow tape from April to June, red tape from July to September and blue tape from October to December.
 - 13.1.2. Tape strips shall be placed one next to the other as follows: January will have the first green strip of tape, February will have a second green strip of tape next to the first one and March will have a third strip of green tape next to the second one.
 - 13.1.3. Inspections shall be done by a Competent Person and in case of electrical tools and equipment by a qualify electrician.

14.

Short Service Employees

- 14.1. Contractor shall implement a Short Service Employee (SSE) Management Program to ensure employees, who are new to the Contractor organization, are appropriately identified, trained, supervised, and developed. This program would target, track, promote, and monitor the continued development of each new employee's safe working habits during the first sixty (60) days of employment.
- 14.2. Contractor's SSE program shall focus on new Contractor or Subcontractor workers (i.e., less than 60 days with the Contractor or Subcontractor organizations) and shall be in effect for each new Contractor or Subcontractor employee for a period no less than 60 days.
- 14.3. Contractor's SSE program shall address and require the following, as a minimum:
 - 14.3.1. Unique identification (e.g., different color hard hat or hi-visibility vest) for new Contractor or Subcontractor employees.
 - 14.3.2. New Contractor or Subcontractor employees shall not be permitted to work alone during the employee's first 60 days.
 - 14.3.3. New Contractor or Subcontractor employees shall be included in the safety committee and/or safety walk down functions on a rotating basis for the first 60 days.
 - 14.3.4. Contractor shall conduct additional H&S training on a monthly basis, at a minimum, for new Contractor and Subcontractor employees in the program.
 - 14.3.5. Contractor shall implement a review process before releasing new Contractor or Subcontractor employees from the SSE program.

15.

Site Access and Security

- 15.1. Unless otherwise indicated by Siemens Energy, Contractor shall develop a specific Site Security Plan including workers access, visitor access, access routes and security procedures.
- 15.2. Firearms, alcohol, and illegal drugs are prohibited on the Project Site. This includes storage in personal vehicles. Subject to any applicable state or local laws, any vehicle, personal toolboxes, and lunchboxes entering the Project Site are subject to search.
- 15.3. Contractor shall ensure that its personnel do not consume alcohol or illicit drugs or are under the influence of the same at Site. Any person found in possession and/or under the influence of alcohol or illicit drugs at Site will be immediately removed and shall not be permitted to return.
- 15.4. Any person involved in incidences of criminal or violent acts shall be immediately removed from Site and not be permitted to return.
- 15.5. Siemens Energy may conduct alcohol/drug tests at Site at any time.
- 15.6. Smoking shall only be permitted in areas designated for that purpose.
- 15.7. Persons temporarily on site for a meeting, tour, or other non-employment activity are typically escorted by assigned Project personnel. Personnel without a card or "Valid" status in the database as applicable may be allowed "Escorted Access" to a site with Siemens Energy approval. However, they are prohibited from performing any hands-

on work (i.e., physical / manual labor).

16. Zero Harm Workshops

- 16.1. Contractor and Subcontractors must prepare and be available to attend a series of workshops that are intended to strengthen the project Zero Harm Culture through understanding and collaboration. These workshops will begin before site mobilization and spread over the course of the Project.
- 16.2. Contractor must coordinate to make available the necessary personnel requested by Siemens Energy to attend the workshops.
- 16.3. Siemens Energy reserves the right to set the location for these workshops and is not responsible for the costs of Contractor attendance and participation.

17.

Zero Tolerance Policy

- 17.1. Contractors shall implement a Project specific Zero Tolerance Policy that, at a minimum, includes provisions for:
 - 17.1.1. Utilize, fall protection measures where a fall of 1.8 m or 6 feet could occur.
 - 17.1.2. Verify scaffolds have been properly inspected / tagged before use. Modifications can only be done by a Qualified scaffold erector.
 - 17.1.3. Obtain an approved Confined Space Permit before entering a Permit Required Confined Space.
 - 17.1.4. Ensure excavations 1.6 m or 5 feet are effectively sloped/ shored/ benched before entry.
 - 17.1.5. Follow all Permit to Work requirements, including Lock Out Tag Out.
 - 17.1.6. Report all injuries and incidents supervision immediately.
 - 17.1.7. Safeguard when working at height, that tools/equipment are secured against dropping and the area below is barricaded preventing access.
 - 17.1.8. Use of cell phone or tablet (talking or texting) while operating construction equipment or machinery is prohibited.
 - 17.1.9. Ensure safety documentation is completed accurately and honestly
 - 17.1.10. Refrain from misconduct which can affect a worker's personal safety.
 - 17.1.11. The use of personal entertainment devices, including headphones, are prohibited on site.
- 17.2. The workers and/or visitors will sign a form acknowledging their understanding and acceptance of the Zero Tolerance Rules and that a violation will result in dismissal from the Project Site
- 17.3. Contractor shall conspicuously post signs or banners displaying the Zero Tolerance Behaviors at the workers / visitor's entrance area to the site and lunch/break area where applicable.

NEEDS SIGNS/POSTERS MADE

SAFETY:

18. Abrasive Blasting / Spray Coating

- 18.1. Abrasive Blasting
- 18.2. Inspect abrasive blasting equipment before use. Check the static pressure drop of the exhaust ventilation system for the abrasive blasting room or booth. Refer to manufacturer's instructions.
- 18.3. Frequently check exhaust ventilation filters. Replace per manufacturer's instructions.
- 18.4. Assure dust does not accumulate on the floor and aisles outside the blasting enclosure. Keep walkways clear of steel shot or similar abrasive, which may create a slipping hazard.
- 18.5. Use signs and/or ribbons to designate temporary blasting areas.
- 18.6. Before removing blasting hood, brush, dust or vacuum abrasives or other contaminants from hood and cape.
- 18.7. At the end of the work shift, shower and change clothes when necessary to remove contaminants.
- 18.8. Before performing work to prevent build-up of static electricity, assure blast hose is grounded to the pot.
- 18.9. Assure blast equipment is maintained by Qualified persons familiar with the hazards.
- 18.10. Use the following personal protective equipment (PPE): work gloves, hearing protection, safety glasses, NIOSH-approved air-supplied blast hoods with capes or aprons.
- 18.11. Spray Coatings
- 18.12. Be familiar with the Safety Data Sheet (SDS) for spray coating materials.
- 18.13. Assure ventilation is adequate or wear respiratory protection (NIOSH-approved).
- 18.14. Before each use, check the static pressure drop of the exhaust ventilation system for the spray room or booth. Follow manufacturer's instructions.
- 18.15. Designate temporary spray-painting areas by using signs or warning tape.
- 18.16. Eliminate ignition sources in spray painting areas.
- 18.17. Inspect spray painting equipment before use.
- 18.18. At the end of the work shift, shower and change clothing when necessary to remove contaminants.
- 18.19. Assure airless spray guns that atomize paints and fluids at high pressures (1,000 pounds or more per square inch) are equipped with automatic or visible manual safety devices to prevent release of the paint or fluid until the safety device is manually released.
- 18.20. Use the following personal protective equipment: eye protection, respiratory protection

as necessary and solvent-resistant gloves.

19. Bucket Trucks and Digger Derricks

- 19.1. Contractor using bucket trucks and or digger derricks must have written procedures for the safe use and operation.
- 19.2. Users/operators shall be Qualified to use the equipment in accordance with applicable standards and regulations.
- 19.3. Manufacturer's operating instruction manual be available onsite
- 19.4. Documented daily inspections of the equipment shall be conducted by a Qualified person and maintained on site.
- 19.5. Wear fall protection in the form of a full body harness and lanyard attached to the manufacturer's prescribed anchorage point.

20. Chain Saws

- 20.1. Follow manufacturer's instructions for operation and adjustment.
- 20.2. Wear PPE including hard hat, goggles or face shield with safety glasses, hearing protection, hand protection and foot protection.
- 20.3. Wear leg protection (chaps) when using a chain saw while standing on the ground. Does not apply to pole saws.
- 20.4. Chain saws, when not in use, shall be properly protected with a blade guard or case.
- 20.5. Gasoline powered chain saws shall be equipped with a continuous pressure throttle control system that will stop the chain when pressure on the throttle is released.
- 20.6. The chain saw shall be started with the chain brake engaged.
- 20.7. Chain saw shall not be operated with one hand.
- 20.8. Do not cut any material other than wood.
- 20.9. Ensure that the chain saw chain does not contact any obstruction such as other logs or branches, rocks, fencing, nails or spikes.
- 20.10. Avoid contact of the bar tip with any object to reduce kick-back hazards.
- 20.11. The chain saw shall be started at least 10 feet from the fueling area.

21. Compressed Gas Cylinders

- 21.1. Handling and moving gas cylinders
 - 21.1.1. Where removable caps are provided for valve protection, such caps shall be kept on cylinders at all times except when cylinders are in use.

Subject to Management Review and Approval

- 21.1.2. Do not lift cylinders by the cap.
- 21.1.3. Never drop cylinders or permit them to strike against other objects.
- 21.1.4. Avoid dragging or sliding cylinders.
- 21.1.5. Never use cylinders as roller supports or for any purpose other than to contain the content as received.
- 21.1.6. Never handle the cylinder with a lifting magnet. Slings, ropes or chains shall not be used unless provisions have been made on the cylinder for appropriate lifting devices, such as lugs.
- 21.1.7. Use suitable hand trucks, fork truck roll platform or similar device with cylinder firmly secured for transporting and unloading.
- 21.1.8. Open cylinder valve slowly. Point the valve opening away from you and other persons. Avoid the use of a wrench on valves equipped with hand wheels. Never hammer the valve wheel when attempting to open or close a valve.
- 21.1.9. Personal protective barricading facilities shall be provided when transferring gases from one cylinder to another.

21.1.10. Cylinder valves shall be closed before moving cylinders, when work is finished and when cylinders are empty. Before regulator is removed from the cylinder, release all pressure from the regulator.

21.1.11. Empty cylinders shall be marked "Empty" or "MT," segregated from full cylinders and returned to the supplier with all valves closed and valve protection caps in place.

21.2. Storage of Gas Cylinders

21.2.1. Cylinders shall be stored in accordance with SDS or other guidance provided for a specific gas.

21.2.2. Cylinders shall not be stored near highly flammable substances.

21.2.3. Cylinders shall be kept away from radiators or other sources of heat.

21.2.4. Do not store cylinders where they could become part of an energized circuit.

21.2.5. Cylinders shall be stored in assigned places, away from elevators, stairs or gangways. Assigned storage places shall be located where cylinders will not be knocked over or damaged by passing or falling objects.

21.2.6. Cylinders shall not be stored in unventilated enclosures.

21.2.7. Compressed gas cylinders not designed to be used or stored horizontally shall be tightly secured above the midpoint in an upright position at all times except, if necessary, for short periods of time while cylinders are being moved with appropriate equipment.

21.2.8. Cylinders stored on mobile vehicles shall be secured in an upright or horizontal position and guarded to prevent any damage or movement. Regulators and gauges shall be removed from compressed gas cylinders while in transit unless designed for that purpose.

21.2.9. Cylinders shall be protected from rusting.

21.2.10. Cylinders shall be hydrostatically tested according to standards of Compressed Gas Association Inc.

21.2.11. Oxygen cylinders in storage shall be separated from fuel gas cylinders or combustible materials (especially oil or grease) by a minimum distance of 20 feet, or by a non-combustible barrier at least 5 feet high having a fire-resistance rating of at least one-half hour.

21.2.12. Oxygen and acetylene can be stored on dolly cart with regulators attached if the cart has a fire-rated barrier separating the cylinders. Without a fire-rated barrier, store cylinders on carts with regulators attached only if it is anticipated that they will be used within 24 hours.

21.2.13. Compressed air hoses with an inside diameter exceeding ½ inch must have a safety device at the source of supply or branch line to reduce pressure in case of hose failure or disengagement of a connection.

21.2.14. Petroleum-based products shall not be used on any equipment where oxygen passes. Keep valves, torches and regulators of oxygen and fuel gas cylinders free of oils, grease and other combustibles.

22.

Confined Space Entry

22.1. For a space to be considered "confined" it shall meet ALL of the following three criteria:

- 22.1.1. Is large enough for an employee to enter fully and perform the assigned work;
 - 22.1.2. Is not designed for continuous occupancy by the employee; and
 - 22.1.3. Has a limited or restricted means of entry or exit.
- 22.2. For a space to be considered a “permit-required” confined space, must also at least have one or more of the following characteristics:
- 22.2.1. Contains or has the potential to contain a hazardous atmosphere;
Note: In order to determine if a confined space contains a hazardous atmosphere, the Contractor must perform an atmospheric test for potential contaminants, oxygen level and LEL.
 - 22.2.2. Contains a material with the potential to engulf someone who enters the space;
 - 22.2.3. Has an internal configuration that might cause an entrant to be trapped or asphyxiated by inwardly converging walls or by a floor that slopes downward and tapers to a smaller cross section;
 - 22.2.4. And/or contains any other recognized serious safety or health hazards.
- 22.3. If Siemens Energy and Contractor disagree with regard to the classification of a confined space as non-permit required, Siemens Energy reserves the right to require the Contractor to reclassify the space and follow the appropriate permit-required confined space procedures.
- 22.4. Contractor shall have a written Confined Space Entry procedure / program, which is acceptable to Siemens Energy that requires all confined spaces be evaluated by a Qualified person and classified prior to entry. Entries shall be made in accordance with OSHA 29 CFR 1926, Subpart AA.
- 22.5. Contractor shall complete and maintain a Confined Space Register form or similar accepted by Siemens Energy for all confined spaces (e.g., permit-required and non-permit spaces) on the Project site, which are applicable to Contractor's scope of work, unless otherwise approved by Siemens Energy.
- Note: Once the area has been turned over to Siemens Energy for commissioning, Contractor and Subcontractors must adhere to the Siemens Energy Permit to Work program and shall follow the Siemens Energy Confined Space Entry Program when entering spaces under commissioning, unless otherwise accepted by Siemens Energy. In these cases, Siemens Energy will provide the location of each Permit-Required Confined Space, hazards or potential hazards of each Permit-Required space the Contractor will enter, and any historical data available on previous permit-required entries into the applicable confined spaces.
- 22.6. Entry Employer (i.e., company entering the confined space) shall assess, evaluate and classify the confined space at the time of entry by a Qualified person based on the hazards present or the hazards created by the entry.
- 22.6.1. A numbered Confined Space Hazard Assessment form approved by Siemens Energy shall be used to evaluate confined spaces for potential hazards prior to entry, list the required hazard control measures, and identify the person(s) who evaluated and authorized entry into the space.
Entry Employer shall document the basis for determining all hazards in a permit space have been eliminated or controlled through a certification that contains the date, location of the space, and signature of the person making the determination. The certification shall be made available to each person entering the space or to that individual's authorized representative in

accordance with the OSHA standard.

- 22.6.2. A copy of the signed, numbered Confined space hazard assessment form shall be posted outside the confined space at all times for worker review so entrants understand the classification of the space and potential hazards inside.
- 22.7. All confined space entries shall be coordinated through the Contractor. Entries shall be documented / logged by Contractor unless otherwise accepted by Siemens Energy. All entries shall be logged, tracked, monitored, and closed by Contractor as Work is completed and entries are terminated by the use of an Entry Log or similar form accepted by Siemens Energy.
 - 22.7.1. If the Controlling Contractor (e.g., EPC, Prime, or General Contractor) is not on site (i.e., has not yet mobilized or is demobilized), Siemens Energy will coordinate the entries performed by the Subcontractors.
 - 22.7.2. Entry numbers on the Entry Log shall match or coincide with the entry numbers on the related Hazard Evaluation form.
- 22.8. All entrants and attendant(s) shall be briefed on the hazards of the space, symptoms of exposure, required tools, personal protective clothing and equipment, and communication methods. Review of a pre-approved RAMS and documented pre-job brief shall be utilized for this discussion prior to entering the space.
 - 22.8.1. Confined Space Entry training records shall be provided to Siemens Energy for all entrants, attendants, entry supervisors, rescue personnel, and Competent Persons who work inside a confined space or are associated with confined space entries.
 - 22.8.2. Training records shall indicate the date of training and course title / type to indicate each person's level of training.
- 22.9. All Permit-Required Confined Spaces (as defined by OSHA 29 CFR 1926.1202) shall be entered by the form of a permit and atmospheric tested, at a minimum. The Entry Employer is responsible for air monitoring in confined spaces and calibration to manufacturer recommendations. All calibration and bump tests shall be documented. All personnel conducting air monitoring shall have proper, documented training.
 - 22.9.1. Prior to entry, hazards inside the confined space shall be eliminated or minimized to the furthest extent possible by ordinary, customary control methods [e.g., ventilation, hazardous control of energy / lockout tagout (LOTO), fall protection system, etc.].
 - 22.9.2. If all hazards are eliminated, isolated, and adequately controlled (i.e., no actual or potential serious hazard), the confined space shall be reclassified as a Non-Permit Confined Space for as long as the hazards remain eliminated or isolated.
 - 22.9.3. Classification and reclassification of the confined space shall be made by a Qualified person using the confined space hazard assessment approved form clearly documented and posted at the entry to the confined space.
 - 22.9.4. When multiple employers are working in a Permit-Required Confined Space, only one employer (e.g., EPC, Prime, or General Contractor) shall have operational control over the permit space. Under no circumstances are two entry permits allowed to be posted for the same confined space by separate employers.
 - 22.9.5. Entry Employer shall inform the Controlling Contractor of any hazards confronted or created inside a Permit-Required Confined Space. Siemens Energy shall also be debriefed and provided this information from the

Controlling Contractor.

- 22.9.6. Permits for entry into a Permit-Required confined spaces may only be renewed or remain open for a maximum of one work week (i.e., 5 business days or 7 calendar days) depending on Contractor's Project schedule / rotation.
- 22.10. Non-entry rescue is the preferred method for retrieval of entrants from Permit-Required Confined Spaces (as defined by OSHA 29 CFR 1926.1202) or Non-permit required confined spaces with complex or difficult access / egress because trained personnel can immediately begin to retrieve incapacitated entrants without entering the space. Entry Employer is responsible for providing rescue / retrieval equipment at their cost.
- 22.11. Prior to entry into a Permit-Required Confined Space (as defined by OSHA 29 CFR 1926.1202), a Qualified attendant shall be at the space all times while entrants are inside. A pre-planned rescue plan and rescue equipment shall be put in place and rescue services shall be notified and put on stand-by.
 - 22.11.1. Rescue equipment shall be in use [i.e., "non-entry" rescue equipment shall be set up or necessary "entry" rescue gear shall be staged (preferably at the entry portal) for prompt use].
 - 22.11.2. Rescue team (e.g., emergency responders) shall notify the Entry Employer if the rescue service is unavailable to provide rescue service at the time of entry inside a Permit-Required space. Permits may be suspended and entries cancelled or postponed when temporary changes occur (e.g., loss of power or fire department / rescue service becomes temporarily unavailable).
 - 22.11.3. A minimum number of four (4) appropriately equipped rescuers shall be staged with immediate access to Permit-Required Confined Spaces if the potential exists for conditions Immediately Dangerous to Life and Health (IDLH). Rescue / retrieval shall be accomplished within 4 minutes so resuscitation efforts can begin promptly.
 - 22.11.4. Rescue / retrieval shall be accomplished within 30 minutes for Permit-Required entries without a hazardous atmosphere. Entrant(s) shall be retrieved from the space in 15-30 minutes, which includes response time to the space, rescue equipment assembly and setup, and actual rescue of entrant(s).

- 22.11.5. Contractor is responsible for providing rescue services at their cost. Siemens Energy preference is for hazards to be eliminated and controlled with non-entry rescue in place so workers can safely enter confined spaces rather than work inside confined spaces with serious potential hazards that require entry-rescue services.
- 22.12. If a public agency (e.g., local fire department) is used to provide rescue from Permit-Required Confined Spaces, a letter of understanding shall be written between the company and the Agency that details, including but not limited to, training, drills/ practice on confined spaces to be entered, site inspections of confined spaces on the Project, equipment to be maintained by each party, and duties / responsibilities of each party, and drills performed to evaluate the effectiveness of the Agency. A copy of the letter shall be provided to Siemens Energy.

23. Cranes and Rigging

- 23.1. Contractor shall submit a "Crane and Rigging Management Plan" to Siemens Energy for review and acceptance. The plan shall require the following information as a minimum:
- 23.1.1. Designated crane management resources
 - 23.1.2. Roles and responsibilities
 - 23.1.3. Qualifications
 - 23.1.4. Training
 - 23.1.5. Coordination
 - 23.1.6. Communication
 - 23.1.7. Severe weather
 - 23.1.8. Inspections
 - 23.1.9. Lift planning
 - 23.1.10. Compliance
- 23.2. Contractor shall appoint a Qualified Crane Supervisor to ensure Contractor crane and rigging operations adhere to the Crane and Rigging Management Plan requirements. Each Subcontractor on site, who utilizes cranes, shall also appoint a Crane Supervisor.
- 23.3. Crane Supervisors shall have at least 10 years' experience as a certified crane operator or Qualified Rigger and shall possess a current certification from an accredited organization for either crane operation and/or rigging activities. The Crane Supervisor shall have sufficient background and knowledge to develop / evaluate lifting plans, select assign correct crane to work activity, inspect cranes, assess competence of crane operators, and administer a safe Crane and Rigging Program.
- 23.4. Individuals who operate cranes and derricks shall possess certification from the National Commission for the Certification of Crane Operators (NCCCO) or other recognized authority. A copy of their certification(s) shall be submitted to Siemens Energy prior to operating crane equipment on the site.

- 23.4.1. Contractor shall perform crane-specific familiarization examinations for all operators on the project prior to operating the crane. Crane operators shall be evaluated when a different type of crane (e.g., size, manufacturer, or model number) arrives on site and/or they are reassigned to a crane they have not operated before.
- 23.5. Rigging operations shall only be performed by Qualified Riggers. Individuals who perform rigging operations shall be qualified per OSHA Subpart CC.
 - 23.5.1. Training shall include instructional training, practical training, verification of required knowledge either by written or oral examination, and the demonstration of required skills under the observation of an approved evaluator.
 - 23.5.2. The rigger is responsible for inspection and assembly of the rigging hardware and lifting equipment as well as securing the load to the rigging hardware and lifting equipment. All rigging shall be inspected prior to use.
 - 23.5.3. The rigger shall inform the flagger that the load is ready to lift.
 - 23.5.4. The flagger is responsible for ensuring that the load is rigged satisfactorily, either by personal inspection or by verification with the rigger.
 - 23.5.5. The flagger is responsible for directing the lift operation and communication with the lift equipment operator.
 - 23.5.6. The flagger should wear a reflective vest and/or reflective gloves.
 - 23.5.7. Flaggers for mobile cranes shall be Qualified signal persons.
 - 23.5.8. Roller block, hoist, lifting and mechanical pulling hooks shall be fitted with a safety latch or secured so that rigging components remain under control. Hooks designed to have a safety latch shall have the latch installed and, if inoperable or missing, the hook shall not be used.
 - 23.5.9. Crane hooks shall not be painted.
 - 23.5.10. Slings shall be used in accordance with the manufacturer's permanently attached identification markings.
 - 23.5.11. If a rope is used as a sling for rigging and attached to a mechanical device, it shall have a splice in lieu of a knot. Anything being lifted by hand (on a hand line) can be lifted using a rope with a knot as long as the rope (including the knot reduction) is rated for the weight being lifted and it can be done safely.
 - 23.5.12. When wrapping slings around sharp corners or edges, softeners or padding shall be used to prevent damage.
 - 23.5.13. The manufacturer's recommendations for safe loading of ropes, cables, chains and slings shall not be exceeded.
- 23.6. Crane signaling shall only be performed utilizing a Qualified Signal Person. Signal Persons shall be qualified per OSHA 1926.1428 for the type of signaling to be conducted (e.g., radio, hand). Signal Person assessments shall be conducted by a third-party accredited evaluator or Contractor's Qualified evaluator per OSHA 1926.1428 option 01 and 02.
- 23.7. Contractor shall maintain an up-to-date register of all cranes on site that lists the crane manufacturer, type / size, unique identification number (e.g., serial number), and current annual inspection date.

- 23.8. Workers shall operate equipment and conduct pre-operational inspections according to the operator's manual.
- 23.9. Contractor shall implement a permit process to coordinate multiple cranes and/or other boom-type equipment (e.g., aerial lift) sharing "airspace." The permit shall require the review and approval by operators, signal persons, and all involved for affected equipment.
- 23.10. All lifts require planning shall be done in accordance with ASME P30 Planning for Load and Handling Activities. A pre-lift planning session and job briefing shall be held for all personnel involved with the lift.

23.11. Contractors shall have a licensed engineer / PE review and approve all critical lift plans and submit critical lift plans to the Siemens Energy representative for review and acceptance, prior to executing / performing critical lifts.

23.11.1. Critical lifts are defined as (1) any lift that utilizes more than one crane, (2) any lift that weighs more than 50 tons (3) any lift involving a crane suspended work platform, including personnel baskets, (4) any lift over critical operating and/or process equipment (i.e., necessary for continual operation of the facility), (5) any lift that exceeds 75 percent of the crane's load chart (6) any lift within 10 feet of energized equipment, (7) any lift within 50 feet of energized power lines, (8) lifts in winds in excess of 25 mph, (9) lifts utilizing special temporary lifting frames (e.g., gantry, elevated runways, jack and slide operations, strand jacks, etc.), or (10) any crane setup / lift on a floating device, such as dock or barge.

23.11.2. Critical lift plans shall include crucial information and address key components of the lift ranging from method of attachment to the load through to any requirements for securing the load before unhooking. In addition, critical lift plans shall address soil bearing conditions and provide a written narrative describing the sequence of the lift with specific steps that explain key components, such as crane setup, proper installation of attachment points, and any temporary securement requirements prior to releasing load. All relevant and necessary information required by the persons executing the lift [e.g., crane operator(s), riggers, signal person, and essential personnel] shall be included and explained in the written plan.

23.12. Contractors shall develop written lift plans for all complex lifts as defined below. Complex lift plans shall be developed by a licensed engineer or Qualified Rigger prior to complex lifts being made.

23.12.1. Complex lifts are defined as (1) any lift utilizing more than one lifting device (e.g., chain hoists / chain falls, come-a-longs, tuggers, and etc.), (2) any lift involving two points on a monorail, (3) any lift involving transferring a load (i.e., handshake) between any combination of lifting devices (i.e., typically referred to as "bull rigging"), (4) loads with a center of gravity more than 12 inches off of center, (5) loads with lifting points on the bottom of the load, or (6) extremely long loads (e.g., pipe or pipe spools) having the potential to buckle when lifted.

23.13. For overhead bridge and gantry cranes:

23.13.1. Not loaded in excess of its rated load capacity except for test purposes.

23.13.2. These lifts are limited to cranes with load rating of 5 tons and above. Load shall not exceed 125 percent of rated capacity. Frequency shall not exceed two in a continuous 12-month period.

23.13.3. Crane manufacturer shall be consulted if the planned engineered lift exceeds

125 percent of rated load or if frequency exceeds two during a continuous 12-month period.

23.14. Contractors shall be responsible for obtaining and providing to Siemens Energy lift information (e.g., load and material / equipment weight calculations) for materials and equipment from manufacturers that is delivered to the Project site. This includes information regarding lifting lugs, spreader bars, center of gravity, and etc. that are intricate and necessary for safely unloading / loading equipment or materials.

23.15. No personnel shall ride on the crane carriage/platform while a crane is in operation.

23.16. All cranes shall be equipped with a functioning travel limiting, anti-two blocking device.

23.17. All tower cranes shall be equipped with "anti-collision" warning devices.

23.18. All tower cranes shall be erected at a height to prevent collision with other cranes, unless otherwise accepted by the Siemens Energy.

23.19. Cranes shall be inspected annually by a third-party person. Documentation shall be maintained on the crane and a copy submitted to Siemens Energy prior to utilizing the crane.

23.20. Cranes shall be inspected and documented by a Competent Person each shift prior to crane operations.

23.21. Cranes shall be inspected and documented by a Competent Person on a monthly basis.

23.22. Signal persons shall possess a whistle, air-horn, or other signaling device to warn of flying loads.

23.23. All loads shall be thoroughly inspected for loose members, nuts, bolts, tools, banding, wire, and debris prior to making a lift to reduce the potential for falling objects.

23.24. Spotters shall be used at all times during travelling (i.e., moving from one location to another) of cranes.

23.25. Rigging hardware (e.g., shackles, eyebolts, slings, and wire rope) located below the hook shall be certified in compliance with ANSI / ASME standards, and shall display its weight capacity, unless otherwise approved by Siemens Energy.

23.26. If ANSI / ASME standards certified spreaders and other lifting devices are not available in the country market they shall be certified by a licensed professional engineer and the capacity shall be marked on the device. This includes design, inspection, and testing per the requirements of ANSI / ASME B30.20.

23.27. Outriggers shall be extended according to the manufacturer's load charts.

23.28. Cribbing shall be used as needed to level the crane and assure stability based on soil class.

23.29. Crane setup location shall be established to assure safe lifting capabilities and avoidance of obstacles.

23.30. Loads shall be guided by tag lines and not by pulling on the rigging.

- 23.31. Tag lines shall be of sufficient length that will allow for proper control without placing worker under the load.
- 23.32. Softeners, constructed of substantial and sufficient materials, shall be used on sharp edges and shall be of adequate size / capacity and shall be inspected before each use.
- 23.33. Slings and rigging equipment shall be inspected by a Competent Person on a monthly basis, at a minimum, and included in the Contractor's color-coding inspection program. Documentation shall be maintained of all monthly inspections, including item identification, condition, date, and inspector.
- 23.34. All rigging shall be in closed configuration (i.e., no shake out hooks, plate clamps/dogs, power magnets, etc.) unless otherwise accepted by Siemens Energy.
- 23.35. Safety latches shall be on all hooks. "J" hooks shall not be used.
- 23.36. All rigging shall be stored properly on racks or shelves to prevent damage and housekeeping issues.
- 23.37. When work is completed at any given location, the boom shall be lowered, retracted and secured and outriggers fully retracted before moving.
- 23.38. Using Cranes to lift (hoist) individuals:
- 23.38.1. The use of equipment to hoist employees is **prohibited** except where the employer demonstrates that the erection, use, and dismantling of conventional means of reaching the work area, such as a personnel hoist, ladder, stairway, aerial lift, elevating work platform, or scaffold, would be more hazardous, or is not possible because of the project's structural design or worksite conditions.
 - 23.38.2. Contractor shall notify Siemens Energy management in writing at least 14 days prior to the intended use of crane mounted personnel platform.
 - 23.38.3. Test lifts shall be performed prior to each shift when utilizing suspended personnel platforms. Test lifts shall be conducted at every location where the suspended personnel platform will be utilized.
 - 23.38.4. Siemens Energy reserves the right to approve or deny the use of crane mounted personnel platforms.
- 23.39. Nighttime operations
- 23.39.1. Nighttime operations shall be conducted with adequate illumination of the pickup area, set down area, the flagger, the boom tip and the load path (if the load path is not previously determined to be clear).
 - 23.39.2. Nighttime crane operations where contact with energized electrical lines is possible shall not occur unless such lines can be adequately illuminated, de-energized or protected.
 - 23.39.3. The crane operator shall determine the required illumination needed for safe operation.
- 23.40. Power line safety
- 23.40.1. For operations near overhead power lines, a mobile crane operator must comply with these rules. This section does not apply to work on transmission and distribution lines and equipment.
 - 23.40.2. Define the work zone.
 - 23.40.3. Make a power line hazard assessment.
 - 23.40.4. If any part of crane can get closer than trigger distance (20 feet for lines under

350 kilovolts (kV); 50 feet for lines over 350 kV) take additional steps. Select Option 1, 2 or 3:

- a) Option 1: Have the lines de-energized and visibly grounded at the worksite.
- b) Option 2: Maintain minimum clearance distance of 20/50 feet AND implement the “encroachment prevention measures.”
- c) Option 3: Maintain distances in Table 1 below. Determine the line’s voltage by asking the utility AND implement the “encroachment prevention measures.”

23.40.5.If Option 2 or 3 is selected, implement these “encroachment prevention measures”:

- a) Conduct a planning meeting with the operator and other workers who will be in the area.
- b) If used, tag lines must be nonconductive.
- c) Erect elevated warning lines, barricades or line of signs.
- d) In addition to the three measures above, also select one of these measures:
 - Use a proximity alarm.
 - Use dedicated spotter (must be a Qualified signal person).
 - Use a device that automatically warns the operator to stop (range control warning device).
 - Use device that limits range of movement.
 - Use an insulating link/device between the end of the load line and the load.

Table 1: Required Clearances for Operations near Overhead Power Lines

Normal Voltage, kV (phase-to-phase)	Minimum Required Clearance (feet)
0 to 50	10
Over 50 to 200	15
Over 200 to 350	20
Over 350 to 500	25
Over 500 to 750	35

Note: Minimum required clearance is 10 feet plus 4 inches for every 10 kV over 50 kV.

23.41.Cranes traveling under power lines

23.41.1.Cranes traveling under a power line with no load must comply with these rules. This does not apply to T&D crane operations.

23.41.2.The boom/mast and boom/mast support system must be lowered sufficiently to meet the requirements below.

23.41.3.The clearances specified in Table 2 below must be maintained.

23.41.4.The effects of speed and terrain on crane movement (including movement of the boom/mast) are considered so that those effects do not cause the minimum clearance distances specified in Table 2 to be breached.

23.41.5.If any part of the equipment while traveling will get closer than 20 feet of the power line, the employer shall assure that a dedicated spotter who is in continuous contact with the operator is used. The dedicated spotter shall:

- a) Be positioned to effectively gauge the clearance distance.
- b) Where necessary, use equipment that enables the dedicated spotter to communicate directly with the operator.
- c) Give timely information to the operator so that the required clearance distance can be maintained.
- d) Be a Qualified signal person, completing signal person training and written and practical test.

23.41.6.When traveling at night, or in conditions of poor visibility, the operator shall assure that:

- a) The power lines are illuminated, or another means of identifying the location of the lines shall be used.
- b) A safe path of travel is identified and used.

Table 2: Minimum Clearance Distances While Traveling With No Load and Boom/Mast Lowered

Voltage (nominal, kV, alternating current)	While Traveling Minimum Clearance Distance (feet)
Up to 0.75	4 (while traveling/boom lowered)
Over .75 to 50	6 (while traveling/boom lowered)
Over 50 to 345	10(whiletraveling/boomlowered)
Over 345 to 750	16(whiletraveling/boomlowered)
Over 750 to 1,000	20(whiletraveling/boomlowered)

23.42.Inclement weather mobile crane operation

23.42.1.When a local storm warning has been issued, a Competent Person must determine if it is necessary to implement manufacturer's recommendations for securing the equipment.

23.42.2.For adverse weather conditions, observe operating wind velocity limits of the equipment based on manufacturer's recommendations that consider wind loads on the surface area of the crane and the lifted load.

23.42.3.When inclement weather is imminent or anticipated, develop a lift plan and operating procedures that include termination of lifting activities at specified conditions, anchoring procedures, the lowering and tying down of boom using cribbing and the requirements for attaching load line to suitable anchors.

23.42.4.Discontinue operations when lightning is present and when visibility is insufficient.

24.

Cutting, Grinding and

Welding Operations

24.1. Contractor's incident and illness prevention plan shall include welding, grinding, and

cutting operations health hazards such as potential exposure to lead (lead paint), arsenic, hexavalent chromium components, asbestos laden primer, and other known harmful substances arising from hot work operations.

24.2. Contractor shall implement a Hot Work Permit process for all hot work (e.g., cutting, welding grinding, etc.), unless otherwise accepted by Siemens Energy.

24.3. The Contractor shall ensure that a suitable fire extinguisher is available at each hot work location.

24.4. The Contractor shall ensure that falling and flying sparks are contained, and Site equipment is protected by the use of fire-resistant screens and/or mats.

24.5. The Contractor shall ensure that protective screens are installed to prevent welding flash injuries to Site personnel.

24.6. Mobile fuel gas supply facilities shall be set up at the workplace or as near to it as practicable and should where possible be in the employee's field of vision, without creating any supplementary risks.

24.7. Gas cylinders and welding equipment must be accompanied by conformity and certification documents and/or labels.

24.8. Liquefied Petroleum Gas and acetylene fuel gas supply facilities must be safeguarded against flashback.

24.9. All gas cylinders, full or empty, shall be secured in the upright position without risk of falling over.

24.10. Empty gas cylinders shall be removed from the work area as soon as practicable.

24.11. The Contractor shall ensure that in the case of electric welding, strict attention must be paid to ensure that the earth clamp is connected to the object to be welded at a point as near as practicable to the weld and never be connected to any other component parts.

24.12. When electric arc welding or oxygen-arc cutting is performed in locals where the walls or sides can conduct electricity, and conditions are cramped and mobility limited, The Contractor shall ensure that DC welding generator sets with no-load voltages not exceeding 100 V may only be used. AC welding generator sets are only permissible when their no-load voltage, for frequencies up to 50 Hz, does not exceed 42 V (42 V marking). Welding rectifiers may only be used if they bear the appropriate identification mark.

24.13. A dedicated Fire Watch with an appropriate fire extinguisher shall be utilized in work areas with combustible materials and/or where the fire, slag, or sparks cannot be contained at all times. A fire watch shall continuously monitor the work area while hot work is performed and at least 30 minutes following the completion of work.

24.13.1. Fire Watchers shall be trained in order to perform the duties of a Fire Watch. At minimum, they shall be trained on the roles and responsibilities, requirements, recognition of hazards, control methods, and fire extinguisher use.

24.13.2.All affected walkways, travel paths, and etc., beneath overhead hot work areas/operations shall be identified with the appropriate barricade tape with completed signage posted at all possible access points.

24.13.3.All affected floor areas, grating, floor openings, etc., beneath hot work areas shall be covered to prevent slag and sparks from falling to areas below.

25. Diving and Water Operations

- 25.1. The following requirements are as well applicable to when working on barges, watercraft and boats.
- 25.2. Employees working over or near water, where the danger of drowning exists, shall be provided with U.S. Coast Guard-approved life jacket or buoyant work vests.
- 25.3. If the drowning hazard can be completely removed through the use of 100 percent fall protection (without exception), life jackets/vests are optional.
- 25.4. Contractor shall consider fall hazards and surrounding structures for all personnel working within six (6) feet or, over, structure over or submerged in water.
- 25.5. No one shall work alone when working in, over, or near water. At least two (2) personnel shall be present at all times working under a "buddy system" and buddies shall never be more than 10 feet apart.
- 25.6. When fall protection is used and will prevent persons from falling into water, personal flotation device is not required.
- 25.7. Prior to and after each use, personal flotation devices shall be inspected for defects that would alter their strength or buoyancy. Defective units shall not be used.
- 25.8. Ring buoys with at least 90 feet of line shall be provided and readily available for emergency rescue operations. Distance between ring buoys shall not exceed 200 feet.
- 25.9. At least one lifesaving skiff shall be immediately available at locations where workers are working over or adjacent to water.
- 25.10.All lifesaving boats or skiffs shall be equipped with a method of propulsion (i.e., motor) effective for the water conditions. Also, paddles or oars shall always be available.
- 25.11.Assure operators of equipment (e.g., cranes, drills) are familiar with the equipment and can demonstrate the skills necessary to safely operate it from an unstable platform (e.g., barges, dredges).
- 25.12.Assure operators have demonstrated skills and/or training to operate watercraft safely.
- 25.13.If mobile equipment is operated near or in water over 30 inches deep, a stand-by person shall be present for the duration of the operation and a Water Rescue Plan shall be in place. The stand-by person shall be identified to all personnel involved in the operation and listed on the pre-job brief form.
- 25.14.For diving operations, Contractor shall submit a Dive Plan for review and comment by Siemens Energy. Dive plans shall be developed and implemented by the dive Contractor for each separate diving operation.
- 25.15.Dive plans shall comply with OSHA 29 CFR Part 1910, Subpart T - Commercial Diving

Operations and Association of Diving Contractors International (ADCI), International Consensus Standards for Commercial Diving and Underwater Operations. Dive plans shall include but are not limited to a pre-dive hazard assessment, diver equipment checklist, dive area assessment, pre-dive briefing, and requirements for cold water diving.

- 25.16. Contractor shall utilize hypothermic protection (i.e., cold water PPE), such as dry suits, anti-exposure coveralls / work suits, survival suits, offshore float suits, and etc., for personnel working in water or on water (i.e., boats, vessels, floating structures) where there is a risk of falling into the water and hypothermia is a potential hazard due atmospheric and/or water temperatures.
- 25.17. When combustible materials are present on barges, watercrafts and boats, assure fire extinguishers are readily available.
- 25.18. Assure safe access/egress to/from any platform or barge.
- 25.19. Designate and maintain unobstructed walkways/aisles with anti-skid material.
- 25.20. Assure handrails are installed unless their presence is a more significant safety hazard.
- 25.21. Assure all barge connections are in place before operation (e.g., barge- to-barge, spud wells-to-barge and anchorage).
- 25.22. Assure engineering has approved all barge modifications.
- 25.23. Safely secure all material/equipment to the barge deck.
- 25.24. Assure watercraft longer than 16 feet are equipped with a Coast Guard- approved throwable personal flotation device.

26.

Dropped

Object

Prevention Program

- 26.1. Contractor shall implement a Dropped Object Prevention Program (DOPP) prior to working at heights greater than 6 feet. Contractor's DOPP program shall be submitted to Siemens Energy for review and acceptance at least 30 days prior to working at elevations >6 feet. Contractor shall address equipment / material / tool security and dropped object prevention system(s) and specific control requirements for the following work locations / surfaces (at a minimum).
 - 26.1.1. Elevated work platforms (e.g., grated catwalks, scaffolds, and overhead working surfaces >6 feet above lower elevations)
 - 26.1.2. Aerial lifts (e.g., scissor lifts, articulating boom lifts, and vertical lifts / pickers)
 - 26.1.3. Suspended work platforms (e.g., suspended scaffolds, Spider® baskets, Sky Climber® swing stages, and crane-suspended personnel baskets)
 - 26.1.4. Ladders (e.g., A-frame / step ladders, extension ladders, fixed ladders, and temporary ladders)
- 26.2. Contractor shall secure all tools and materials stored on elevated work platforms.

26.3. Contractors shall provide perimeter netting on the guardrails (toe board to top rail) on elevated platforms (including aerial lifts) and scaffolds where the potential for falling objects exists. Perimeter and debris netting shall be a fine mesh with openings ½-inch or less designed to be installed vertically or horizontally to prevent objects from falling to lower levels.

26.4. Contractor shall red barricade the drop zone below elevated work areas and place signage to prevent employees from entering where the potential for falling objects exists. Contractor shall install debris netting or canopies below elevated work areas where the potential for falling tools, materials, and/or objects exists when barricading the area below or tethering is not feasible.

26.5. Floor openings and holes shall be covered to prevent tools, equipment, and materials from falling through holes, gaps, voids, and crevices. Gaps, voids, and/or holes (i.e., two inches or more in its smallest dimension) in any floor, roof, or other elevated walking / working surface through which an object may fall >6 feet to a lower elevation shall be covered.

26.6. Contractor shall secure all tools with tool tethers when working at heights greater than six (6) feet. Tool tethers shall be a manufactured product that is designed and utilized for its intended use. Makeshift tethers utilizing rope, duct tape, or electrical tape are prohibited.

26.7. When working around energized equipment, tethers shall be constructed of non-conductive materials.

26.8. Tool tether installation (e.g., attachments, D-rings, shrink wrapping) shall only be performed by trained personnel.

26.9. Tools weighing more than 5 pounds shall not be tethered to the body. Heavy tools should be attached to a suitable structure, anchor, or support so the shock load from a dropped tool does not affect the worker's balance causing them to fall and/or injure body parts (e.g., soft tissue injuries, strains, sprains, and etc.).

26.10. Contractor shall use appropriately sized, secured tool bags to lift tools. Manufactured ropes with built-in or pre-connected locking carabineers or locking snap hooks shall be used to lift tools in tool bags. Knots shall not be used to secure ropes to tool bags. (Tools that could potentially fall from the tool bag during transport shall be tethered to the bag.)

26.11. Buckets shall not be utilized for tool storage and transport to elevated work areas.

26.12. Rope may be utilized to hoist building materials and equipment to elevated work areas. Rope for hoisting and lowering materials and equipment shall be of proper size, material type, and free of defects.

26.12.1. These ropes shall be fastened by incorporating a locking carabineer or rated hitch, splice, braid, cinched loop, or knot (e.g., figure 8 on a bight, bowline knot, slip knot, etc.) by a skilled person, such as a Rigger, so the connection cannot become undone inadvertently.

26.12.2. Ropes shall be properly choked or cinched to prevent shifting or displacement of materials while lifting or lowering.

- 26.13. Contractor shall train all personnel on their Dropped Object Prevention Program. A dynamic learning experience / practical application (i.e., hands-on) of dropped object prevention measures (e.g., tool tethers, floor covers, and debris netting) shall be provided as part of the training.
- 26.14. Contractor shall conduct and document weekly dropped / falling object hazard assessments, audits, and/or inspections when worked is performed at heights >6 feet. Contractor shall provide to Siemens Energy within 5 business days the findings of these evaluations, summary of trends or issues take appropriate actions to prevent future deficiencies or issues from occurring, track follow-up / action items through to completion, and report the status of these back to Siemens Energy.

27.

Electrical Safety and

Energy Control

27.1. General Requirements

- 27.1.1. Contractors shall have a written Lockout / Tagout Program that address sources of uncontrolled energy that have the potential to cause personal injury. The program shall identify hazardous energy sources associated with construction temporary electrical installations, work activities, locking and tagging procedures at isolation boundaries, and verification of zero energy.
- 27.1.2. Contractor's site-specific safety orientation shall include Lockout / Tagout (LOTO) "Affected Employee" training for all employees, Subcontractors, vendors, and visitors who work in an area where LOTO activities may occur.
- 27.1.3. Contractors may be subject to Siemens Energy control of hazardous energy procedures / LOTO program and shall abide by the provisions.
- 27.1.4. Only Authorized Employees shall de-energize and lock / tag machines and equipment. Before beginning work on de-energized systems and/or equipment, employees shall verify by testing or operation that hazardous energy has been relieved.
- 27.1.5. For electrical isolations where employees may contact conductors that were previously energized, a Qualified person shall use a voltage detector to test the circuit elements and electrical parts to verify they are de-energized. Voltage detectors shall be checked for proper operation as described in its operations manual immediately before and immediately after use (i.e., live-dead-live test).
- 27.1.6. Contractor shall be responsible for legibly labelling all breaker panels, switch boxes, disconnects, and etc. used by the Contractor and Subcontractors with the respective voltage and phase rating and what the box / panel / service controls.
- 27.1.7. Contractor shall be responsible for legibly labelling / tagging all temporary systems and equipment installed by the Contractor and Subcontractors with the appropriate equipment and system information along with contact information of the installer and end user.
- 27.1.8. Contractor shall utilize tags and labelling that are weather resistant and suitable for the environment in which they will be used so that information remains legible.

- 27.1.9. When zip ties / cable ties are used to attach various LOTO tags (e.g., lockout tags, test tags, custody tags, and etc.), the zip ties shall have a tensile strength of at least 250 pounds and have ultra-violet (UV) protection.
- 27.1.10. Contractor shall ensure light plants, generators, vacuum trucks / equipment, fuel trucks, and other mobile equipment are set up and grounded per the manufacturer's guidelines.
- 27.1.11. Immediately at the beginning of commissioning activities, contractor shall follow Siemens Energy permit to work process and submit formal requests to Siemens Energy prior to taking a mechanical or electrical system out of its design configuration once it is installed where there is hazard or risk of an energy release (e.g., pneumatic, hydraulic, stored / potential, electrical, mechanical, etc.).
- 27.2. High Voltage Testing
- 27.2.1. Only skilled, trained, and experienced electricians / test technicians may perform AC / DC High Voltage Testing (e.g., Double Testing, Hi-Pot, Megger, etc.). Testing shall be performed in accordance with the manufacturer's directions and a detailed written test plan and a RAMS shall be done as part of commissioning activities, reviewed by Siemens Energy prior to performing this work. In addition, the below control measures shall be followed and addressed in the RAMS and Pre-Job meeting:
- 27.2.2. Contractor shall establish a Controlled Work Access Zone around test equipment / apparatus and equipment being tested. Use temporary barricades (e.g., red tape and signs) to mark or delineate these boundaries.
- Unqualified electrical workers and/or workers without adequate PPE are required to stay outside the Minimum Approach Distances for the voltages being tested. Ensure proper PPE is utilized during testing and/or when working on or near exposed energized parts.
 - Testing devices shall be grounded according to manufacturer instructions before making test connections on the equipment to be tested.
 - Test trailers or test vehicles shall have their chassis grounded before testing. If not, the test set shall be moved out of the trailer or van to ensure proper grounding of test equipment.
 - Only Qualified electrical workers shall install and remove grounds. A lone worker may operate a bonding disconnect or ground switch, but this switch shall not be used for personal protective grounding unless it is specifically designed for the purpose of grounding.
 - Before installing grounds, ensure proper isolation / clearance of conductors and equipment. Isolation points shall be identified by a review of the schematic diagram(s) as a part of the overall testing plan. Verify isolation / clearance from all voltage sources by means of visible open points according to switching instructions and test for voltage with an approved voltage detector.
 - Equipment shall be de-energized, isolated from the system, tested for voltage, and personal protective grounds applied before testing. Personal protective grounding can be removed for testing. This requirement shall be identified and addressed in the test procedure.
 - Ungrounded parts of the equipment or test leads shall not be touched

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- until the absence of voltage is confirmed with an approved voltage detector.
- If the cable or equipment is normally energized with primary voltage, the cable or equipment shall be grounded before installing test equipment. Remove grounds as needed when applying test voltage.
 - Cables and conductors, which are not under test that are in contact with the equipment being tested, shall also be grounded.
 - Additional administrative controls shall be used when the attachment of grounds to certain cables (e.g., DCS, signal, feedback, low voltage, etc.) is not feasible. These administrative controls shall be included in the test plan and submitted to the Siemens Energy for review and acceptance.
- g. If bystanders could come in contact with or come near a cable under test, the test equipment or any other exposed conductors associated with the test, the following extra protective safety measures must be taken:
- Cordon off the area with red barricade tape and signs where the test is being performed.
 - Monitor the test area to keep bystanders out of danger.
- h. The safety ground connection shall be applied to the test set before making any other connections. Always connect leads to the test set prior to applying them to the equipment.
- i. Class 2 rubber gloves or a live-line tool shall be used while handling test leads once the test leads are installed into the test set. The Operator's Assistant shall wear Class 2 rubber gloves or use a live-line tool while connecting and disconnecting test leads on the equipment being tested.
- Workers shall not hold test leads to support them or make contact with test leads while they are energized during testing. Workers shall secure test leads so they remain under positive control at all times during testing.
- j. No other work activities shall be performed on any circuits associated with the component being tested during the testing period.
- k. After performing any voltage-inducing test, the voltage switch on the test set shall be turned to the OFF position. Personnel shall verify the voltage detector drops to zero. Equipment shall be grounded to remove any capacitive charge. If possible, personnel shall leave a ground on the bushings or terminals that are not being tested to prevent voltage build-up.
- l. Contractor shall utilize equipment as designed (i.e., do not modify switches). Safety switches or safety features shall not be bypassed, disabled, or modified.
- m. Only the Test Set Operator and the person "hanging the hook / clamp" (Operator's Assistant) shall have the arming devices. It is not permissible to delegate the arming device to another person.
- n. Test Set Operator shall establish and maintain visual contact with Operator's Assistant before the test begins. Radio contact alone between the Test Set Operator and the Operator's Assistant does not

Subject to Management Review and Approval

meet the requirement for visual (i.e., face-to-face) contact with these two individuals. Radio contact can be used to enhance communication.

- If line of sight is not physically possible (e.g., extended distances, physical barriers / obstacles, or elevation changes), a third person (Designated Safety Observer) shall be used to provide verification to the Test Set Operator that the Operator's Assistant is free and clear of equipment prior to arming the test set and relay information back to the Test Set Operator if conditions change.
 - Designated Safety Observer shall maintain constant face-to-face contact with the Operator's Assistant and relay verification back to Test Set Operator that the Operator's Assistant is not in contact with test leads or parts of equipment that could become energized while testing is in progress.
 - As long as the Designated Safety Observer is in constant face-to-face contact with the Operator's Assistant, radio communication between the Safety Observer and Test Set Operator can be used to verify/confirm the Operator's Assistant is in a safe location. Without a third person observer in place, radio contact alone between two parties (e.g., Test Operator and Operator's Assistant) is inadequate.
- o. Contractor shall use clear 3-way communication technique (i.e., Repeat Back Method) before energizing testing equipment and again when testing is complete.
 - p. After testing is complete, Qualified electrical workers shall de-energize the circuit that was tested, reinstall protective grounds, and perform zero energy verification prior to re-entering the electrical enclosure and/or working on the equipment. Lockout / Tagout (LOTO) procedures may require additional steps or boundaries in order enter electrical enclosures.

27.3. Energized Switchyard / Substations

27.4. Access to energized Switchyard / Substation shall only be done by Qualified commissioning personnel as a part of the commissioning plan, and at a minimum the following requirements shall be followed for areas outside defined substations walkways:

- 27.4.1. Prior to entering an energized switchyard / substation, a detailed Pre-Job meeting shall be performed to discuss work activities to be performed, potential hazards, and hazard controls. Even if working alone, the Pre-Job meeting shall be documented and readily available (e.g., posted or on person).
- 27.4.2. Individuals shall wear 100% natural fiber clothing and flame-resistant (FR) clothing with an arc rating of at least 4.2 Cal/cm² when performing work on energized substation equipment or when within 10 feet of exposed energized substation equipment.
- 27.4.3. Garments worn shall not have any screen prints, iron-on prints, or other ink prints unless the garment is made of all natural fibers or FR material, this also applies to any garments worn under FR clothing.
- 27.4.4. In addition to toe-protection, protective footwear shall also have electrical hazard (EH) rated soles that cover the entire foot with no openings. Approved safety shoes shall meet the requirements associated with ASTM F-2413,

"Standard Specification for Performance Requirements for Protective Footwear."

- 27.4.5. All jewelry shall be removed prior to entry. This includes, but is not limited to finger rings, necklaces, earrings, body piercings, chains, wrist bands, wrist watches, key rings / chains, metal or plastic bracelets, or studs.
- a) Prescription safety glasses with side shields and medic alert necklaces, bracelets, and shoe tags are not considered jewelry and are exempt from this requirement but shall be secured to prevent potential contact, flash, or catching hazards.
 - b) Medic alert necklaces and bracelets shall be worn under FR clothing or rubber protective gloves if working on energized conductors or other energized equipment.
- 27.4.6. Individuals requesting unescorted physical access to energized substations and/or switchyards shall be trained at a minimum on:
- a) Trained in and familiar with any other safety practices, including applicable emergency procedures (such as pole-top and manhole rescue),
 - b) The skills and techniques necessary to distinguish exposed live parts from other parts of electric equipment.
 - c) The minimum approach distances specified in this section corresponding to the voltages to which the Qualified employee will be exposed and the skills and techniques necessary to maintain those distances
 - d) The proper use of the special precautionary techniques, personal protective equipment, insulating and shielding materials, and insulated tools for working on or near exposed energized parts of electric equipment, and
 - e) The recognition of electrical hazards to which the employee may be exposed and the skills and techniques necessary to control or avoid these hazards.

All this training in accordance with OSHA 29 CFR 1910.269(a)(2)(i-ii) and/or as required by the Siemens Energy and shall abide by any additional client procedures or provisions not addressed in the document.

28.

(EWP)

Elevating work platforms

28.1. General

- 28.1.1. This section outlines requirements for the operation of boom-supported elevating work platforms, self-propelled elevating work platforms and manually propelled elevating work platforms (aerial and scissor lifts).
- 28.1.2. The operator shall assure the operating and maintenance manuals are available.
- 28.1.3. The operator shall comply with manufacturer's warnings and instructions.

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- 28.1.4. Personal fall arrest equipment is required for workers working from boom-supported elevating work platforms.
- 28.1.5. **Workers working in scissor lifts shall be protected by guardrails and personal fall arrest.**
- 28.2. Pre-start equipment inspection
 - 28.2.1. Before use each day, a documented pre-start inspection shall be conducted in accordance with operator's manual.
 - 28.2.2. Any problems or malfunctions shall be repaired prior to use of the platform.
 - 28.2.3. Outriggers are used as required
 - 28.2.4. Guardrails are installed and access gates closed
 - 28.2.5. Load and its distribution on the platform are in accordance with manufacturer's rated capacity
 - 28.2.6. All personnel on the platform have PPE for the work and environment envisioned
- 28.3. Workplace inspection
 - 28.3.1. Before the elevating work platform is used and during use, the operator shall check the area for possible hazards including:
 - 28.3.2. Drop-offs, holes and slopes
 - 28.3.3. Bumps and floor obstructions
 - 28.3.4. Debris
 - 28.3.5. Overhead obstructions and electrical conductors
 - 28.3.6. Hazardous atmospheres
 - 28.3.7. Inadequate surface and support to withstand load forces imposed by the aerial platform
 - 28.3.8. Wind and weather conditions
 - 28.3.9. Presence of unauthorized persons
 - 28.3.10. Other possible unsafe conditions
- 28.4. Driving requirements
 - 28.4.1. Maintain a clear view of the support surface and route of travel.
 - 28.4.2. Assure personnel in the worksite area that may be affected are aware of the movement, communicating and maneuvering the platform as required to protect against personal injury.
 - 28.4.3. Maintain a safe distance from obstacles, debris, drop-offs, holes, depressions, ramps and other hazards to assure safe travel.
 - 28.4.4. Maintain a safe distance from overhead obstacles.
 - 28.4.5. Maintain the minimum approach distance from energized power lines. For operators who are not qualified transmission and distribution (T&D) line technicians, the distances are:
 - a. For lines under 50 kV – 20 feet
 - b. For lines over 50 kV – 20 feet plus 4 inches for every 10 kV over 50 kV

29.

Equipment, Vehicle, and

Tool Inspection

- 29.1. Daily equipment / vehicle inspection – Operators shall inspect all safety and operational features on a pre-shift basis. Operators shall visually check for leaks and/or signs of spilled fuel, oil, hydraulic fluid, grease, coolant, and etc. during this inspection. Equipment operators shall check equipment for proper operation as per manufacturer's safe operating manual or guidance. This applies to any piece of equipment or vehicle including cranes.
- 29.2. Fire extinguishers inspections shall be conducted monthly (at a minimum) and all fire extinguishers shall have inspection tags and annual maintenance record (e.g., tag, label). Fire extinguishers shall be inspected and maintained in accordance with OSHA 1926.150 and NFPA 10A. Maintenance, servicing, and recharging of portable fire extinguishers shall be performed by trained and certified persons.
- 29.3. The following equipment and materials shall be uniquely identified and inspected monthly, by a Competent Person and marked to show the current inspection color tape according to color-coding system, and the inspections shall be documented. Items found defective shall be removed from site or destroyed. Items included in the inspection program includes, but is not limited to:
 - Electrical cords, GFCI's, electric tools
 - Ladders
 - Fall protection equipment
 - Rigging
- 29.4. Retractable lifelines shall be inspected by a third-party Qualified person on an annual basis per manufacturer guidelines. A copy of inspection records shall be maintained at the Project site.
- 29.5. Contractor shall provide to Siemens Energy the list of color codes for safety inspections. Color coding system for the Project shall be mutually agreed upon by Contractor and Siemens Energy.
- 29.6. Construction elevators, or equivalent, shall be inspected by a third-party Qualified persons after erection and annually thereafter.

30.

Excavations and Ground

Penetrations

- 30.1. Prior to commencement of construction, Contractor shall self-perform or have a third party perform a "pre-construction" underground survey. The survey shall be conducted by utilizing a combination of the following methodologies: ground penetrating radar (GPR) technology, electromagnetic (EM) testing equipment that induces an electrical current into the earth, conductive testing of trace wires, site walk down, research of historical information and review of drawings, and/or aerial photographs. The methods (e.g., geophysical prospecting equipment) used to conduct the pre-construction underground survey will be determined by onsite conditions with the final decision made by Siemens Energy. The underground survey shall be conducted for the Contractor's

construction "limits of disturbance" with the completed survey / drawings / mapping provided to Siemens Energy.

30.2. On "green field" sites that have not been previously developed (e.g., industrial, commercial, or residential), GPR and EM testing shall be conducted around the entire perimeter of the Project site. Any underground utilities identified by GPR, EM testing, or locate services (e.g., One-Call or Call Before You Dig) shall be fully traced, physically marked, and documented (e.g., GPS location and depth). In addition, GPR and EM testing shall be performed over all utility corridors, surface depressions, or other suspect areas or anomalies identified in aerial photographs, historical research, and/or site walk downs.

30.3. Due to the uses and limitations with varying types of locating equipment, unpredictable conditions, and numerous considerations on project sites, a joint collaboration between Siemens Energy and Contractor will be made regarding the "pre-construction" underground survey plan with the final decision made by the Siemens Energy.

30.4. Utility location devices to be used shall be capable of detecting utilities and underground structures both conductively and inductively (i.e., metallic and non-conductive materials). Survey information shall include both vertical (depth) and horizontal plane / location of the underground utilities and subsurface structures.

30.5. Any contractor whose scope includes trenching, excavation, or ground penetration activity shall submit Trench and Excavation Permits, subject to Siemens Energy review and approval, identifying the excavation area, potential hazards, protective measures, and amount of time (e.g., hours, days, or weeks) the permit will remain open. Trench and Excavation Permits shall be posted in a visible location at the work location and/or a copy shall be maintained in the respective excavating equipment. All personnel involved in the trenching and excavation, or ground penetration shall be briefed on the permit requirements and the job prior to starting work.

Trench & Excavation Permits shall only remain open for a defined period of time as identified on the permit. Permits shall not remain open indefinitely and shall be closed / terminated as the task or work activity is completed.

30.6. Ground penetration or "breaking ground" is defined as any time the earth surface is penetrated (e.g., excavation, trenching, auger drilling, grading, post driving, hand digging, etc.). Anytime the earth's surface is penetrated, the Trench and Excavation Permit process shall be utilized prior to breaking or penetrating the ground regardless of depth.

30.7. As part of the Trench and Excavation Permit process, Contractor and Subcontractors are required to review the "pre-construction" underground survey and any underground design drawing(s) prior to breaking ground.

If a discrepancy exists between the available data, it shall be investigated and reconciled by uncovering (i.e., visually locating) the subsurface item via "soft digging" methods to confirm the physical location prior to acceptance of the permit by Siemens Energy and use of mechanical digging, drilling, boring, or driving equipment.

30.8. Contractor shall perform a "pre-dig" underground utility locate test utilizing an underground locating device, such as an electromagnetic testing device, just prior to breaking ground to either verify the absence of any utility / structure or confirm the location of previously identified utilities in the pre-construction survey. Any variance to this requirement will be made by Siemens Energy on a case-by-case basis.

30.8.1. The "pre-dig" test above is not the same as the "pre-construction" underground

survey, which covers the Project site. This "pre-dig test" is in addition to the "pre-construction" underground survey and is used to validate/ confirm the results of the initial survey.

- 30.8.2. The "pre-dig" test is performed specifically within the excavation corridor or area where the ground will be penetrated just prior (i.e., <3 working days) to performing the Work. The pre-dig test may be conducted by Contractor employee, who is trained, or a third party.
- 30.8.3. A "pre-dig" test using an underground locating device is not required for surface conditioning that does not penetrate the earth's surface, placement of gravel for roadways, setting of survey flags and wooden stakes, and installation of coir logs, silt sock stakes, and landscape staples. Contractor shall submit to the Siemens Energy for acceptance, a Trench and Excavation Permit, RAMS, and documented field walk down inspection to ensure no underground utilities exist in the area prior to performing these types of work activities.

30.9. Contractor shall provide and maintain above ground identification (e.g., color coded post markers for the applicable utility) for high hazard utilities (e.g., gas, electricity, fiber optic, pressurized lines) as agreed upon with Siemens Energy.

30.10. Prior to breaking ground, a Competent Person shall mark and establish the excavation boundaries and mark / identify hold points at least 10 feet from any underground utility. Hold points shall be clearly marked on the ground (e.g., paint) indicating the presence of a utility. No mechanical excavation activity shall take place within the established hold points until the area has been surveyed with an underground locating device and the utility has been visually located using "soft digging" methods.

30.10.1. Where utilities are identified, Contractor shall "soft dig" (e.g., hydro excavate, vacuum excavate, pneumatic / air blast excavate, or systematically hand dig) to "pothole" and visually locate (daylight) the utility or underground structure, unless otherwise approved by Siemens Energy. Again, mechanical digging shall not take place within 10 feet of an identified utility, unless otherwise accepted by Siemens Energy.

30.10.2. Once soft digging methods expose the buried utility, soft digging methods shall be used to clear / remove earthen material from the utility in order to create a buffer zone of at least 3 feet from the underground line before mechanical digging may resume in the area.

30.11. Earthmoving operations / excavation equipment (e.g., track excavators) shall observe and comply with minimum approach distances based on the voltages to energized overhead power lines. Additional controls (e.g., dedicated Spotter, elevated warning line or barricade, etc.) shall be used if minimum approach distances can be encroached upon by the equipment.

30.11.1. Excavation equipment shall also be grounded when digging near overhead or buried power lines where there is a potential to come in contact with energized conductors.

30.12. Contractor shall appoint a Competent Person to fill out Trench and Excavation Permits and monitor all excavation or ground penetration work.

30.12.1. Prior to personnel entering an excavation each day, it shall be inspected by a Competent Person. Excavations shall be inspected following rainstorms or other occurrences / events that would increase the potential for hazards to develop.

- 30.12.2. Inspections shall be documented and readily available (e.g., posted, stored, or kept inside the operator's cab on equipment) at the excavation location along with the Trench and Excavation Permit.
- 30.13. Each employee working at the edge of a vertical excavation six (6) feet or more in depth shall be protected from falling. Fall protection is not required for excavations, which are sloped or benched, where the horizontal to vertical ratio is less than or equal to ¾:1 ($\leq 53^\circ$ angle). Only rigid barricading will be used at areas where a fall hazard is present, unless otherwise approved by the Siemens Energy. Trench and excavation barricades shall be identified with signs and/or warning tags noting the Contractor's name and contact information.
- 30.14. All open excavations (regardless of depth) shall be barricaded using yellow caution tape.
- 30.15. Tabulated data for engineered, manufactured, or fabricated protective systems (e.g., trench boxes, shields, and shoring) shall be provided to the Siemens upon their request.
- 30.16. Construction excavation or earth moving activities that encounter suspect contaminated soils shall be stopped and the Siemens notified immediately. If an archaeological item is encountered, the work shall be stopped and the Siemens Energy notified immediately.
- 30.17. Trench and Excavation Permits for ground penetrations at or near (i.e., within 100 feet) an existing water impoundment (e.g., dike, dam, levee, lake, pond, earthen basin, reservoir, or canal), wetlands, or of any "Water of the United States" shall be submitted to Siemens Energy prior to penetrating the ground or disturbing the soil in these locations along with documentation verifying environmental permits have been reviewed and required controls are in place.

31. Earthwork Operations

- 31.1. Prior to performing earth moving operations (e.g., timbering / logging, clearing & grubbing, dirt work / cut & fill operations, drilling, blasting, and/or rock breaking), Contractor shall submit to Siemens Energy, for acceptance, a plan that addresses specific control methods and expectations for earth moving equipment operators, truck drivers, and workers to ensure pedestrians are segregated and protected from construction vehicles and mobile equipment while it is moving.
- 31.2. Contractor shall establish Exclusion Zones where earth moving equipment is operated and in motion. Exclusion Zones shall be clearly defined (e.g., signs at each entry point, traffic cones, traffic barrels, or hard / soft barricades) so non-essential personnel can distinctly identify the boundaries of the Exclusion Zone and stay out of the work area.
- 31.3. In areas where earth moving equipment is being operated, Exclusion Zones shall be utilized to eliminate or minimize the need for Spotters and keep foot traffic (i.e., pedestrians) out of the area.
- 31.4. Non-essential personnel shall stop and get permission each time they enter an Exclusion Zone. Prior to granting access, the person in charge of the Exclusion Zone shall alert the earth moving equipment operators in the area.
- 31.5. Mobile equipment operators shall stop if an unauthorized vehicle or person enters the Exclusion Zone without permission, communication is broken with the Spotter, or a pedestrian leaves a Safe Zone.

- 31.6. Before exiting a vehicle inside an Exclusion Zone, all personnel shall remain inside their vehicle and ensure moving equipment within 50 feet comes to a complete stop, throttles or powers down, and the blades, buckets, or other attachments (i.e., implements) are lowered to the ground.
 - 31.6.1. Equipment operators and personnel shall stay in the cab of their equipment or vehicles 100% of the time during tree felling operations.
 - 31.6.2. All personnel are prohibited from entering on foot an area where tree felling is occurring. A minimum distance of two (2) tree lengths of the trees being cut / felled shall be maintained until the machine operator or saw man has acknowledged it is safe to enter the Exclusion Zone.
- 31.7. Inside an Exclusion Zone, no pedestrians are permitted on the ground within 50 feet of earth moving equipment while it is in motion unless a Safe Zone is clearly established with barricades.
 - 31.7.1. Spotters may be used inside Exclusion Zones as an interim or temporary control measure (i.e., <15 minutes) while Safe Zones are being established. While Safe Zones are being established, this is the only time Spotters may be used inside an Exclusion Zone while moving equipment is in operation.
 - 31.7.2. At a minimum, Spotters shall maintain at all times a distance of at least two (2) vehicle lengths from the earth moving equipment while it is in motion. Only one piece of earth moving equipment shall be permitted to operate within 100 feet of a Spotter whenever a Spotter is utilized to protect pedestrian(s) on the ground.
 - 31.7.3. At a minimum, barricades shall provide adequate separation of personnel from moving equipment, and they shall be at least two vehicle lengths in size based on the largest piece of earth moving equipment in the vicinity.
 - 31.7.4. Hard or rigid barricades (e.g., "Jersey" barriers, temporary handrails, or A-frame traffic horses) shall be used for extended activities / operations inside Exclusion Zones for better protection and physical separation from moving equipment.
 - 31.7.5. Inside Exclusion Zones, parked / stationary equipment and vehicles shall be positioned in between pedestrian(s) and the direction of mobile equipment operation, as practical, to provide a physical barrier from the moving equipment.
- 31.8. An Exclusion Zone of 300 feet shall be established in all directions around tub grinders and 200 feet in front of belt grinders. Only designated grinding operations personnel are allowed inside the Exclusion Zone while the equipment is in operation.
- 31.9. At minimum, Red "Danger" or "Do Not Enter" tape or ribbon shall be used to identify and delineate the Exclusion Zone on all sides around drilling and rock breaking operations. Exclusion Zones shall be adequately sized to account for moving equipment and flying debris. Only designated drilling operations personnel are allowed inside the Exclusion Zone while drilling equipment is being operated.

32.

Trucks

Haul Trucks / Delivery

- 32.1. When loading and unloading trucks with motorized mobile equipment (e.g., forklifts,

telehandlers, and cranes), Exclusion Zones shall be established by the use of barricades or traffic cones. Exclusion Zones shall extend 50 feet on each side of the truck trailer (and 25 feet on rear if loaded or unloaded from trailer's rear end) and shall include the total length of the trailer being loaded or unloaded.

- 32.1.1. If a distance of 50 feet cannot be established due to interferences or the presence of permanent structures, equipment, or other physical barriers, these structures may be used as a hard or physical barricade. However, control measures shall be in place to ensure pedestrians do not inadvertently enter the pinch/crush hazard area between the truck and stationary object.
- 32.2. Only material handling equipment operators are permitted inside the Exclusion Zone while trucks are being unloaded with mobile equipment. No movement of material handling equipment is to occur within the Exclusion Area if pedestrians are inside the restricted area.
- 32.3. If any pedestrian needs to enter the Exclusion Zone for any reason the steps below must be followed:
 - 32.3.1. All material handling equipment shall stop, and loading / unloading operations shall cease.
 - 32.3.2. Personnel shall not enter the Exclusion Zone until the material handling equipment operator grants permission.
 - 32.3.3. Operation of material handling equipment (e.g., forklift, telescoping handler, crane, etc.) shall not resume until all pedestrians have exited the Exclusion Zone and returned to the Safe Zone.
- 32.4. Workers shall not remove chains, straps, or banding from a load until the truck has reached its final unloading location (i.e., offload destination on site). Prior to removing straps, chains, and/or banding, the load shall be thoroughly examined to determine/assess if the load has shifted, binders have been broken, stability of the load has been altered, or it is otherwise hazardous to personnel.
- 32.5. Truck drivers shall stay inside their vehicle 100% of the time during truck loading, unloading, and dumping operations. No loads shall travel over the cab. If that is not possible, the driver shall exit the cab and remain inside an established Safe Zone.
- 32.6. Applicable signage, hand signals, radio voice commands, and/or a "Stop & Go" attendant shall be used to control and direct haul trucks in and out of the area where truck dumping / loading activities are performed.
 - 32.6.1. Two-way radios and/or equipment horns shall be used to direct trucks in order to eliminate the need for a Spotter or Flagger on the ground around heavy equipment.
 - 32.6.2. Alternate haul routes shall be considered, as practical, to eliminate or minimize backing up / reversing operations of haul trucks.
- 32.7. Haul trucks are prohibited from dumping material side-by-side. Haul trucks shall be separated at least 1.5 times the length of the truck in all directions.
- 32.8. After a truck has dumped material, it shall travel no further than one truck length until the bed is completely lowered and secured.
- 32.9. Haul trucks that are required to enter a public roadway shall be free of any loose debris that has the potential to fall off and enter the roadway. Tire washes and/or truck washes shall be performed in designed wash bays / basins, as approved by the Siemens

Energy.

- 32.10. Truck beds shall be covered (i.e., tarped) while transporting material and prior to exiting the site. Trucks shall be equipped with a mechanized tarping system or other acceptable method to prevent drivers from climbing trucks where they are potentially exposed to a fall 6 feet or more.

33.

Fall Protection

- 33.1. Contractor shall develop a Fall Protection Plan and submit to the Siemens Energy for review and acceptance.
- 33.2. Fall protection is required when Contractor and Subcontractor personnel are exposed to a fall in excess of 6 feet or when required by additional rules. One hundred percent fall protection is required whether the employee is traveling from Point A to Point B, connecting structural steel, or erecting scaffolds or other temporary platforms.
- 33.3. No personnel or work operation is exempt from the 100 percent fall protection requirement. Controlled Access Zones are not permitted, unless otherwise accepted by the Siemens Energy.
- 33.4. Fall protection anchors shall be independent of other use. Fall protection equipment and anchoring devices shall be exclusively used for fall protection and shall be clearly identified by color coding, tags, and etc.
- 33.5. Any removal of grating or guardrail shall be done under a permit and shall be approved by Siemens Energy. Note that utilizing red barricade tape (e.g., soft barricade) around the entrance or restricted access to an open hole or unprotected edge does not provide fall protection. Rigid barricades shall be utilized around fall protection hazards >6 feet with 100 percent fall protection for workers inside the rigid barricade.
- 33.6. Where employees are required to access work areas beyond rigid barricades, swing gates shall be installed and used on guardrail systems.
- 33.7. Fall protection training shall be provided by a recognized training provider or employer audited program, as agreed upon with Siemens Energy. Refresher training shall be completed every year thereafter.
- 33.8. Acceptable fall protection training shall consist of classroom instruction, including donning the harness, inspection, anchor points, etc. Contractor orientation or OSHA 10 and 30 hour training shall not be accepted as fall protection training.

34.

Protection

Fire Prevention &

- 34.1. General
- 34.1.1. Contractor shall develop a fire prevention and protection program designed to mitigate potential fire risks and identify appropriate firefighting measures, including the following.
- 34.1.2. Contractor shall provide and maintain suitable firefighting equipment adapted to the specific risk in its work areas in accordance with Applicable Laws, and that its personnel receive appropriate training in the use of this equipment.

- 34.1.3. A fire extinguisher rated not less than 10: ABC shall be provided for each 3,000 square-feet of the protected building area or major fraction thereof. Travel distance from any point of the protected area to the nearest fire extinguisher shall not exceed 100-feet.
 - 34.1.4. In multi-story buildings one or more fire extinguishers rated not less than 10: ABC shall be provided on each floor located adjacent to each stairway.
 - 34.1.5. Where more than 5 gallons of flammable or combustible liquids or 5 pounds of flammable gas are being used, a fire extinguisher rated not less than 10: ABC shall be provided within 50-feet.
 - 34.1.6. When in presence of explosive atmospheres all electrical wiring and equipment for light, heat, or power purposes shall be intrinsically safe and or spark free of the type approved for the hazardous locations when applicable.
 - 34.1.7. Internal combustion engine powered equipment shall be so located that the exhausts are well away from combustible materials.
 - 34.1.8. Smoking is prohibited at or in the vicinity of operations which constitute a fire hazard and shall be conspicuously posted: "No Smoking or Open Flame."
 - 34.1.9. The nozzle of air, inert gas, and steam lines or hoses, when used in the cleaning or ventilation of tanks and vessels that contain hazardous concentrations of flammable gases or vapors shall be bonded to the tank or vessel shell. Bonding devices shall not be attached or detached in hazardous concentrations of flammable gases or vapors.
 - 34.1.10. Safety cans 5 gallons or less of flammable / combustible liquids shall be equipped with flame arrestors.
 - 34.1.11. Flammable/combustible material designated areas and approved cabinets shall be labeled "Flammable – No Smoking or Open Flames Within 25 Feet." The quantity of flammable / combustible liquids in any work area shall not exceed the amount required for one day, one shift, or 120 gallons.
 - 34.1.12. Fire extinguishers - All fire extinguisher stations shall be plainly marked and unobstructed for quick access. All fire extinguishers shall be properly mounted and marked for class / type.
 - 34.1.13. Halon-based fire extinguishers and systems are not to be used on Siemens Energy Sites.
 - 34.1.14. Fires in or near electrical equipment (Class C fires), specifically in or around the Combustion Turbine and Steam Turbine Electrical Generators shall ONLY be extinguished by using appropriately sized CO₂ fire extinguishers. Use of any other multi-purpose dry chemical or dry powder extinguishers is not recommended for these areas due to irreversible product damage.
 - 34.1.15. CO₂ extinguishers shall be conspicuously located in these areas and appropriate signage shall be displayed describing this requirement. The Contractor must ensure that employees using CO₂ extinguishers are properly trained in their correct and safe use. At all times, personnel shall act in a safe manner so as not to endanger themselves or other employees.
- 34.2. Temporary buildings
- 34.2.1. Not more than 15 gallons of flammable or combustible liquids shall be stored in a room outside of an approved storage cabinet.

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- 34.2.2. Quantities of flammable and combustible liquid in excess of 15 gallons shall be stored in an NFPA and Underwriter's Laboratory approved metal storage cabinet.
- 34.2.3. Not more than 60 gallons of flammable or 120 gallons of combustible liquids shall be stored in any one storage cabinet, and not more than three such cabinets may be located in a single storage area.
- 34.2.4. Temporary buildings, when located within another building or structure, shall be of either non-combustible construction or of combustible construction having a fire resistance of not less than 1 hour.
- 34.2.5. Temporary buildings, located other than inside another building and not used for the storage, handling, or use of flammable or combustible liquids, flammable gases, explosives, or blasting agents, or similar hazardous occupancies, shall be located at a distance of not less than 10 feet from another building or structure. Groups of temporary buildings, not exceeding 2,000 square feet in aggregate, shall, for the purposes of this part, be considered a single temporary building.
- 34.2.6. At least two portable fire extinguishers having a rating of not less than 20: ABC shall be located outside of the building, but not more than 10 feet from the door opening into any room used for storage of 60 or more gallons of flammable or combustible liquids.
- 34.3. Open yard storage general requirements
 - 34.3.1. Combustible materials shall be piled with due regard stability of piles in no case higher than 15ft.
 - 34.3.2. Driveways between and around combustible storage piles shall be at least 15ft wide.
 - 34.3.3. Driveways shall be so spaced that a maximum grid system unit of 50 feet by 150 feet is produced.
 - 34.3.4. The entire storage site shall be kept free from accumulation of unnecessary combustible materials. Weeds and grass shall be kept down and a regular procedure provided for the periodic clean-up of the entire area.
 - 34.3.5. No combustible materials shall be stored outdoor within 10 feet of a building or structure.
 - 34.3.6. Storage of containers (not more than 60 gallons each) shall not exceed 1,100 gallons in any one pile or area. Piles or groups of containers shall be separated by a 5-foot clearance. Piles or groups of containers shall not be placed within 20-feet of a building.
 - 34.3.7. The storage area is to be curbed or bermed to contain 100% of the contents of the stored liquid in event of a spill.
 - 34.3.8. The storage area and berm are to be lined with an impermeable membrane to prevent contamination of soil by spilled material.
 - 34.3.9. When curbs or dikes are used, provisions shall be made for draining off accumulations of ground water or spills of flammable or combustible liquids. Drains shall terminate at a safe location.
 - 34.3.10. Two or more portable tanks grouped together having a combined capacity in excess of 2,200 gallons shall be separated by a 5-foot clear area.

34.4. Fire Protection Safety Devices

- 34.4.1. Portable heaters, including salamanders, are to be equipped with an approved automatic device to shut off the flow of gas to the main burner and pilot, if used, in the event of flame failure. Heaters having inputs exceeding 50,000 BTU per hour must be equipped with a pilot or an electrical ignition system.
- 34.4.2. Every container and every vaporizer shall be provided with one or more approved safety relief valves or devices. These valves shall be arranged to afford free vent to outer air.
- 34.4.3. Shutoff valves shall not be installed between the safety relief device and the LPG container.
- 34.4.4. Containers, regulating equipment, manifolds, pipe, tubing, and hose is to be located to minimize exposure to high temperatures or physical damage.

35. Gas blow

- 35.1. CAUTION / WARNING: Explosions have occurred while cleaning contractors were using natural gas to clean balance of plant piping. As a result, the U.S. Chemical Safety Board has recommended that Siemens Energy and other manufacturers warn users not to clean balance of plant piping with fuel gas. For further information, see www.csb.gov. Based upon this position, gas blows are prohibited on the Project.
- 35.2. Should Siemens Energy elect to perform natural gas blows against the recommendations, all Siemens Energy personnel will be removed from the site during gas blow activities?

36. Hand and Power Tools

- 36.1. Contractor shall use only tools and equipment appropriately suited for the job. Tools and equipment shall be used in accordance with manufacturer's recommendations as designed for their intended use.
- 36.2. No modification shall be made to tools and equipment without written permission from the manufacturer. Written permission shall be provided to Siemens Energy upon their request.
- 36.3. No hand-made / shop-made tools or equipment shall be used without written permission from Siemens Energy and shall be designed and/or approved by a licensed, professional engineer and manufactured according to the design specification.
- 36.4. Contractor shall inspect tools and equipment before use. Worn, defective, or unsafe tools and equipment shall be immediately removed from service and properly marked / identified until repairs can be made by a Qualified person.
- 36.5. When using a screwdriver or any tool with puncture capability, do not hold work in the hand in such a manner that slipping will cause injury.
- 36.6. Tools shall not be painted, taped or modified in any way that would prohibit inspection for visual cracks or defects.
- 36.7. Electric power tools shall be double-insulated type or electrical cords shall have a ground fault circuit interrupter (GFCI) at the power source.

- 36.8. All air hoses exceeding ½-inch inside diameter shall have a safety device (e.g., excess flow / check valve) at the source of supply or branch line to reduce pressure in case of hose failure or disengagement of a connection.
- 36.9. Hose safety whip checks shall be utilized on all pneumatic air hose connections.
- 36.10. Documentation regarding certified training and manufacturer's recommendations for using Hilti, Ramset, and other powder-actuated tools [OSHA 1926.302(e)], shall be available on site.
- 36.11. All utility knives shall be self-retracting. Use of folding razor knives, folding pocketknives, and spring-assisted / quick opening pocketknives are prohibited on site.
- 36.11.1. Scissoring-action tools (e.g., side cutters / snips, industrial scissors, pliers, or similar tools) shall be used to cut tie wraps / zip ties, non-metallic hoses/ tubing, and etc.
- 36.11.2. Utility knives shall not be used for "skinning" wire or cable. The use of electrician plier-type / scissor-action tools (e.g., wire / cable strippers, cutters, and crimpers) is recommended. Wire / cable skinning knife (i.e., lineman knife with fixed curved blade) may be used to remove insulation jackets on electrical wire / cable. These knives shall be sheathed when not in use.
- 36.11.3. Long blade knives commonly used by insulators are acceptable for cutting insulation only. Insulation knives (i.e., fixed / straight blade) shall be sheathed when not in use by Insulators.
- 36.11.4. Use only box cutters with self-retracting or enclosed blades for opening boxes.
- 36.12. Coverings on the handles of tools shall not be considered as electrical insulation or as a substitute for voltage-rated gloves.
- 36.13. Chisels, punches, drifts and similar tools shall be kept free of mushroomed heads to prevent flying fragments.
- 36.14. Nonconductive measuring devices shall be used when taking measurements on or near exposed energized electrical conductors or equipment.
- 36.15. Handles shall be installed on files and rasps before use.
- 36.16. Open-end wrenches shall be used with force in the proper direction. Do not overload wrenches by using extensions. Ensure good footing when using wrenches to prevent slipping.
- 36.17. When using a striking face (slugging) wrench, use caution. The wrench, when being struck by a hammer, shall not be held by the hand or foot. Use a properly tied rope or other practical means to secure the wrench.
- 36.18. Do not throw tools between other workers or locations. Use a tool bucket or other suitable container and handline to raise or lower tools between elevations.
- 36.19. Portable grinders shall be equipped with a non-locking (i.e., dead man) trigger mechanism and shall be equipped with anti-kickback braking mechanism unless otherwise accepted by Siemens Energy.

37.

Job-made stands

- 37.1. Contractors are prohibited from producing stands for piping and equipment unless the stands have been designed and engineered to meet required performance criteria.

Documentation shall be provided to Siemens Energy prior to the use of any job made stands for approval. Job made stands must be properly labelled with the capacity limitations. Siemens Energy reserves the right to disallow any job produced stands for any reason.

38.

Ladders and Scaffolds

- 38.1. Portable ladders used for access to an upper landing, surface, or excavation shall extend a minimum of three (3) feet above the landing surface. Ladders shall be securely placed, held, or tied / lashed to prevent slipping, falling, or displacement while in use.
- 38.2. Ladders shall be Type 1A according to ANSI A14.5-2000, meaning they are extra Heavy-duty with a working load of 300 pounds. Fiberglass ladders that comply with this specification are preferred ladders.
- 38.3. Employees climbing a temporary ladder with a fall exposure greater than 12 feet shall be protected by an approved cage, ladder climbing device, or by the use of a body harness, lanyard, and/or lifeline system.
- 38.4. Open sides and ends of platforms more than six (6) feet above the ground or floor shall have top rails, mid rails, and toe boards. Platform area shall be fully planked or decked with the space in between planks or decking not to exceed one (1) inch.
- 38.5. Employees shall not use a scaffold unless it is inspected and properly tagged.
- 38.6. Scaffolds shall be inspected at the beginning of each shift and after significant events that could affect a scaffold's structural integrity.
- 38.7. Unqualified persons shall not modify scaffolding.
- 38.8. Following the inspection, the scaffolding Competent Person shall affix the appropriate scaffold tag, as follows, to each access point.
 - 38.8.1. Red – Scaffolding is incomplete and cannot be used.
 - 38.8.2. Yellow – Scaffolding is missing one of more components, however, it can be used with special precautions, i.e. fall protection. The Competent Person must list scaffold deficiencies and special precautions on the scaffolding tag.
 - 38.8.3. Green – Scaffold is complete and safe to use.
- 38.9. Scaffolds which are not tagged must not be used until it is properly inspected and tagged by a Competent Person.
- 38.10. Scaffold users shall be trained in the safe use of scaffolding.
- 38.11. Scaffolding erectors shall be trained in the proper scaffolding erection techniques by a Qualified person.
- 38.12. Scaffold users must inspect all scaffolds before each use. Pay attention to footing, planking, guardrails, mesh, connections, weld, rust, ladder condition or other access means.
- 38.13. Use personal fall arrest equipment on supported scaffold platforms higher than 6 feet if guardrails are not installed.
- 38.14. Use personal fall arrest equipment and a guardrail system on single-point and two-point adjustable suspension scaffolds.

- 38.15. Do not use a defective scaffold. Assure a WARNING, DO NOT USE tag is put on a faulty scaffold until it is repaired.
- 38.16. Use the access provided with the scaffold.
- 38.17. Scaffolds using hatch deck with pull down ladders shall have a third horizontal member over the top handrail to provide fall protection.
- 38.18. Do not accumulate too many tools, materials and debris on the scaffold or overload it beyond the rated capacity.
- 38.19. Do not stand on or lay tools, materials and equipment on any scaffold railing.
- 38.20. To prevent tipping, do not load a scaffold unevenly.
- 38.21. During high winds and storms, do not work on an outside scaffold.
- 38.22. Do not work on an ice-coated scaffold.
- 38.23. Assure mobile scaffolds rest on a solid level footing.
- 38.24. If wheels or casters are provided, lock them to prevent accidental movement on scaffolds.
- 38.25. When asked to move mobile scaffolds:
 - 38.25.1. Get help so the pressure of pushing is not concentrated in one point on the scaffold.
 - 38.25.2. Assure scaffold will move across level floors that are also free of obstructions and openings.
 - 38.25.3. Apply force close to (5 feet or less) the base of the scaffold.
 - 38.25.4. All users have exited the scaffold.
- 38.26. Before performing work from a scaffold, identify and safeguard any electrical hazards.
- 38.27. When moving mobile scaffolding, avoid obstructions including piping equipment, instrumentation, electrical lines, etc.
- 38.28. Keep the scaffold's working surface clean and organized at all times.
- 38.29. When the job is completed, remove all items from the scaffold.
- 38.30. Do not erect, use, dismantle, alter or move scaffolds so that they or any conductive material handled on them comes closer to exposed and energized power lines than:
 - 38.30.1. For lines under 300 V – 3 feet
 - 38.30.2. For lines over 300 V under 50 kV – 10 feet
 - 38.30.3. For lines over 50 kV – 10 feet plus 4 inches for every 10 kV over 50 kV
- 38.31. Do not use the frame on welded tubular frame supported scaffolds to access upper levels unless the distance between the integral rungs is 16
- 38.32. $\frac{3}{4}$ inches or less and the rungs are uniformly spaced. Otherwise, use an extension or hook-on ladder.
- 38.33. Use tag lines or equivalent measures to control swinging loads when they are being hoisted onto or near scaffolds or if the loads could contact the scaffold.
- 38.34. Use ladders on scaffolds to increase working level height of workers only on large-area scaffolds.
- 38.35. On suspension scaffolds inspect wire ropes for defects before each work shift and after

Subject to Management Review and Approval

any occurrence that could affect a rope's integrity. Replace ropes if:

- 38.35.1. Any physical damage impairs the function and strength of the rope.
- 38.35.2. Kinks impair the tracking or wrapping of rope around the drum(s) or sheave(s).
- 38.35.3. Six randomly distributed broken wires are in one rope lay, or three broken wires are in one strand in one rope lay.
- 38.35.4. Abrasion, corrosion, scrubbing, flattening or peening causes the loss of more than one-third of the original diameter of the outside wires.
- 38.35.5. Heat damage caused by a torch or any damage caused by contact with electrical wires is evident.
- 38.35.6. The secondary brake has been activated and has engaged the suspension rope during an over-speed condition.

38.36. Assure that wire rope clips on suspension scaffolds are inspected and retightened to the manufacturer's specifications at the start of each work shift.

38.37. Assure the wire ropes on suspension scaffolds are shielded, treated to resist corrosive substances or made of material that is not adversely affected by the substance being used when acids and other corrosive substances are used.

38.38. Shield suspension ropes when performing a heat-producing process.

38.39. Reduce the possibility of welding current arcing through the suspension wire rope when welding is performed on suspension scaffolds by ensuring the following:

- 38.39.1. An insulated thimble is used to attach each suspension wire rope to its hanging support.
- 38.39.2. Excess suspension wire rope and any additional independent lines from grounding are insulated.
- 38.39.3. The suspension wire rope is covered with insulating material extending at least 4 feet above the hoist.
- 38.39.4. The tail line below the hoist is insulated to prevent contact between it and the platform.
- 38.39.5. The portion of the tail line that hangs free below the scaffold is guided and/or restrained so that it does not become grounded.
- 38.39.6. Each hoist is covered with insulated protective covers.
- 38.39.7. In addition to a work lead attachment required by the welding process, a grounding conductor is connected from the scaffold to the structure.
- 38.39.8. Assure that the size of this conductor is at least the size of the welding process work lead, and this conductor is not in series with the welding process of the work piece.
- 38.39.9. If the scaffold grounding lead is disconnected at any time, the welding machine is shut off.

- 38.39.10. An active welding rod or uninsulated welding lead is not allowed to contact the scaffold or its suspension system.

39.

Machine Guarding

- 39.1. Before starting work using powered machinery or tools, assure that guards:
 - 39.1.1. Are in place so as to prevent contact with dangerous, moving parts by any part of the body (e.g., hands, arms) or clothing
 - 39.1.2. Are firmly secured to the machine, if appropriate, and are made of durable material that will withstand the conditions of normal use
 - 39.1.3. Are appropriate for the equipment
- 39.2. Do not create new hazards by having a shear point, jagged edge or an unfinished surface that can cause a laceration.
- 39.3. Report any broken or missing guards to your supervisor. Do not use the tool until repairs have been made.
- 39.4. Before working, perform a pre-job safety assessment to identify potential hazards to the hand.
- 39.5. When working around machinery with rotating or moving parts:
 - 39.5.1. Assure gloves, ties, loose clothing and long hair do not become entangled in equipment.
 - 39.5.2. Tie back long hair.
 - 39.5.3. Button or properly roll up long sleeves, and tuck in shirttails.
 - 39.5.4. Do not wear loose jewelry (e.g., necklaces, chains, cords) outside clothing.
 - 39.5.5. Wear badge straps of a breakaway-type of material

40.

Material handling

- 40.1. If a load is unstable, unbalanced, awkwardly shaped, or prone to roll, slide or shift, the load shall be secured at all times.
- 40.2. Rigging below the tines of a forklift / powered industrial truck / telescoping handler is prohibited unless performed with a manufacturer's approved lifting attachment. The direct attachment or placement of rigging equipment (e.g., slings, shackles, rings, etc.) onto the tines of a powered industrial truck is prohibited.
- 40.3. Manual handling of materials shall not exceed 50 pounds per person, use a two-person system or assisted mechanical device such as; carts, wheelbarrow, dolly, etc.
- 40.4. Pipe stands shall be inspected (e.g., leg bends, cracked welds, metal deformation, damaged adjusting screw or head assembly, and etc.) prior to use to ensure all parts are in safe operating condition. Any pipe stands found to be in a state of disrepair shall be immediately tagged and removed from site.
- 40.5. Pipe stands shall only be utilized on solid and level surfaces.
- 40.6. Pipe stands shall be clearly marked with the safe working load and shall not be loaded beyond the rated capacity.

- 40.7. Pipe stand should be adjusted to the desired work height prior to loading with only fine adjustments for fit-up to be made after loading.
- 40.8. Load shall be centered on jack head.
- 40.9. Loads shall not be pushed or pulled once loaded onto pipe stand unless stand is designed for such movement, path is level and stable, and load is secured to stand in accordance with manufacturer's recommendations.
- 40.10. No materials or equipment shall be stored under energized power lines unless otherwise approved by Siemens Energy.
- 40.11. The use of alternative methods, such as telescopic conveyors, pump trucks, or flowable materials, shall be considered instead of traditional methods (e.g., skid steer loaders, excavators, etc.) when placing or moving earthen material (e.g., soil, sand, gravel, stone, etc.) in congested areas around plant equipment or personnel where the potential to injure nearby workers or damage equipment or structures is significant.

41.

Personal

Protective

Equipment

- 41.1. Hard hats, safety glasses, foot protection, and hi-visibility clothing are required (at a minimum) unless otherwise accepted by Siemens Energy.
- 41.2. Additional PPE may be required depending on the hazards associated with the work or due to Siemens Energy site specific requirements. PPE shall be in compliance with national recognized standards. i.e.: Hard hats ANSI Z89.1-1986, foot protection ANSI Z41.1-1991, and eye / face protection ANSI Z87.1-1989 shall be ANSI compliant. Respirators shall be NIOSH-approved.
 - 41.2.1. Hard hats and eye protection are required to be worn inside mobile equipment with an open cab. Head and eye protection is not required inside a fully enclosed operator's cab.
 - 41.2.2. Hard hats shall be worn with the brim forward unless using a face shield or welding hood for welding, cutting, or grinding purposes.
 - a) When wearing face shields or welding hoods, hard hats may be worn with the brim backwards to accommodate the face shield or hood if the hard hat's suspension is specifically designed for reverse donning.
 - b) Hard hats shall be worn with the brim forward when face shields or welding hoods are not in use.
- 41.3. Employees are required to wear safety boots with steel or composite toed, smooth Leather uppers, (canvas or suede is not allowed), distinct heel, non-slip sole, lace up with minimum 4 inches ankle protection and support in construction areas.
- 41.4. Where warranted by the hazards, protective footwear shall be worn that also includes metatarsal protection.
- 41.5. Goggles or sealable fitting eyewear (e.g., spoggles) shall be worn, in conjunction with a face shield or welding hood, by all employees performing grinding, welding, and/or cutting operations on any material (e.g., metal, wood, plastic, concrete, etc.) with a powered tool.
 - 41.5.1. Protective eyewear with tinted or semi-mirrored lenses (i.e., slight or minimal

tinging) shall only be worn for outside work in bright environments. Protective eyewear with clear lenses (i.e., no tint) shall be worn for work indoors / inside buildings. This does not apply to welders and helpers (e.g., fire watch), who shall use shaded eye protection appropriate for the type of work when welding or cutting is being performed.

41.5.2. Anyone within 10 feet of grinding, welding, and cutting operations shall wear goggles or sealable eyewear (e.g., spoggles).

41.5.3. Goggles or sealable eyewear and face shield are required within 10 feet of concrete discharge (e.g., concrete chutes, shotcrete operations, etc.).

41.5.4. Contractor and Subcontractors shall provide various styles of eye protection and ensure eyewear is correctly fitted (i.e., proper size for face with negligible gaps / voids between frame and skin). Sealable eyewear (e.g., spoggles) shall be form fitted and seal against the face so flying debris and airborne particles cannot enter.

41.6. Goggles or sealable fitting eyewear (e.g., spoggles) shall be worn by employees who are involved in scaffold erection / dismantling, vacuuming, insulating activities, or other dusty operations where airborne particles / debris are generated.

41.7. Welder helmets shall be used during all arc welding or arc cutting operations, excluding submerged arc welding. Safety glasses will be worn at all times.

41.8. Appropriately shaded spectacles with side shields and cup or cover-type goggles may be used for all gas welding and cutting operations including a clear full-face shield with a shade of no less than number 10.

41.9. Transparent shields or goggles may be used for resistance welding or resistance brazing.

41.10. Welding helmets shall be provided with window lenses and filter cover plates. Eye protection for shielded metal arc welding should be not less than number 10.

41.11. Goggles shall be ventilated to prevent fogging of the lenses.

41.12. Lenses shall bear a permanent distinctive marking, by which the shade may be readily identified.

41.13. Helmets and goggles shall be kept clean and in good repair.

41.14. Hearing protection shall be used when noise levels exceed recommended levels or, as a general rule of thumb, when workers cannot carry on a normal conversation within 2-3 feet of each other. See Hearing Conservation for more information

41.15. Gloves and protective sleeves shall be appropriate to the task and used to protect the hands, wrist area, and forearms from sharp edges, abrasion, chemicals, electrical contact, and other hazards as necessary.

41.16. Gloves with proper cut and puncture resistance shall be worn when using utility knives and handling sharp objects / materials (e.g., glass, metal, ceramic, and other sharp edges).

41.16.1. Cut resistance gloves shall have an American Society for Testing and Materials (ASTM) rating of 3 or higher (ASTM F1790) when using knives or handling glass or sheet metal.

41.16.2. Cut resistant sleeves (i.e., wrist and forearm protection) shall be worn when handling larger materials, such as photovoltaic (PV) cells / panels, sheet glass,

or sheet metal, which are a cut and/or slashing hazard.

41.17. Hi-visibility clothing shall be worn at all times unless otherwise accepted by Siemens Energy.

41.17.1. When working near public roads (i.e., in or near the road right-of-way), hi-visibility clothing shall be ANSI / ISEA 107 compliant.

41.17.2. Contractors, who are exposed to heavy construction traffic, shall wear a minimum of ANSI Type / Class 2 High-Visibility Safety Vest or Clothing. Flaggers shall wear ANSI Class 3.

41.18. Saw men shall wear appropriate PPE when using a chain saw (e.g., chaps, cut-resistant and puncture-resistant gloves, logging boots, face shield, safety glasses, and earmuffs) in addition to standard PPE. Cut-resistant gloves shall be used when sharpening saws.

42.

Powered Industrial Trucks

(Forklifts)

42.1. Powered industrial trucks shall be operated in accordance with manufacturer's recommendations.

42.2. Only trained and authorized operators shall be permitted to operate a power lift truck.

42.3. Powered industrial trucks shall be inspected before being placed in service each day or before each shift. Document inspection.

42.4. A powered industrial truck shall not be used when deficiencies are found. The truck shall be repaired before it is placed back into service.

42.5. The rated capacity shall not be exceeded. When these vehicles are equipped with supports, they shall be utilized to prevent tipping.

42.6. When the powered industrial truck is parked or left unattended, the operator shall place the forks flat on the floor, neutralize controls, turn off the engine, set hand brake and remove/secure the key to prevent unauthorized use.

42.7. While loading or unloading material, forks may be left in the raised position if there is an ergonomic concern, and precautions have been discussed as part of the pre-job briefing.

42.8. The operator may leave the seat of a parked and running powered industrial truck only if:

42.8.1. the operator is in line of sight

42.8.2. the operator is within 25 feet of the controls

42.8.3. the powered industrial truck is secured

42.8.4. the forks are flat on the floor

42.8.5. the controls are neutralized

42.8.6. the hand brake is set

42.9. The operator shall be in the driver's seat while operating a powered industrial truck.

42.10. Modifications shall not be made without written approval from the manufacturer.

Subject to Management Review and Approval

42.11. Attachments must be considered as part of the load and used in accordance with manufacturer's recommendation.

42.12. Free rigging (suspending load from the tines with a sling) is prohibited.

- 42.13. Operator shall not allow anyone to stand or pass under the elevated portion of any powered industrial truck.
- 42.14. The operator shall not descend a ramp or grade with the load in front.
- 42.15. Only the driver shall ride on the powered industrial truck, unless an additional seat is provided.
- 42.16. Powered industrial truck operators and passengers shall wear seat belts where provided.
- 42.17. The operator of an order picker or high-lift rider truck shall wear approved fall protection and lanyard. The lanyard shall be attached to an approved overhead member of the platform at a point located above and near the center of the platform.
- 42.18. When forward view is obstructed, the driver shall travel with the load trailing or get assistance.
- 42.19. Operators shall properly stack and balance loads for transporting.
- 42.20. Operators shall travel with forks or load as close to the ground as possible and spread the forks for proper balance and stability of load.
- 42.21. Arms, legs and feet shall remain inside the powered industrial truck when moving.
- 42.22. Hands and arms shall not be extended through openings in the mast area.
- 42.23. The charger shall be turned off before connecting or disconnecting from an electric powered industrial truck.
- 42.24. Before driving or operating a powered industrial truck inside or on a truck or trailer, those trucks or trailers shall be secured and wheels chocked.
- 42.25. Travel speeds shall be dictated by surface conditions.
- 42.26. The horn shall be sounded before going through doorways or around blind corners.
- 42.27. Plan routes to avoid hazards of overhead power lines. Use spotters when necessary.

43.

Signs and Barricades

- 43.1. Contractors are required to attach hazard information tags and name of supervisory personnel responsible for work on all barricade tape areas.
- 43.2. Rigid or hard barricades (e.g., fencing, temporary railing, concrete or water-filled plastic "Jersey" barrier, or traffic sawhorse / A-frame barricade) shall be utilized to protect high hazard areas (e.g., open holes with fall distance >6 feet) unless otherwise accepted by Siemens Energy.
- 43.3. Red "Danger" or "Do Not Enter" tape / ribbon or hard barricades with appropriate signage shall be used to restrict access into medium or high hazard areas (e.g. suspended loads, dropped objects, imminent danger, hot / live electrical work, etc.). Only the work crew and personnel authorized by the person in charge of the restricted area are permitted to access or pass through a red, danger / restricted access barricade.

- 43.4. **Yellow "Caution"** tape / ribbon shall be used to alert personnel to minor hazards (e.g., slip / trip hazards, housekeeping issues, hot surfaces, **open excavations** etc.). Any person may access into a yellow / caution barricade area as long as they have familiarized themselves with the hazards detailed on the **barricade signage** (i.e., hazard information tag).
- 43.5. Contractor shall provide signage specific to the hazard communication needs of both English and non-English-speaking personnel that may be working on the Project site.
- 43.6. Hard barricades (e.g., concrete Jersey barriers) and/or parked vehicles shall be used as shields when personnel are working on the ground along, near, or adjacent to a public roadway where they may be exposed to or struck by moving vehicles.
- 43.7. Office complexes and bathroom facilities that are exposed to construction traffic shall be protected from mobile equipment and vehicular traffic (e.g., traffic control barricade, concrete Jersey barriers, etc.).

44.

Site Installations

- 44.1.1. All temporary underground installations, such as power, water, sewer, etc., shall be installed in compliance with Applicable Codes and Standards and maintained on a set of as-built drawings. At the conclusion of the Project, temporary underground installations shall be handled in one of the following manners:
 - 44.1.2. If installed in accordance with applicable codes and incorporated into permanent plant drawings, the utilities may be de-energized or isolated and left for the Plant's future use.
 - 44.1.3. If utilities or services are determined to be non-compliant with applicable codes, they shall be de-energized or isolated at the source and removed, unless otherwise accepted by Siemens Energy.
- 44.2. The installation and removal of any Site facility is subject to approval by Siemens Energy.
- 44.3. Where Siemens Energy does not provide the terminal points for potable water, wastewater and electric power, Contractor is fully responsible at its own expense for the supply of these services in accordance with Applicable Laws and general standards of health, hygiene and safety.
- 44.4. Responsibilities for the installation of telephone lines and connections and any other communication support services if not at Contractor expense, will be defined contractually.
- 44.5. The Contractors Site facilities shall be reasonably identified with the Contractor's name outside on all 4 walls and with an "Entrance" sign.
- 44.6. Where a Contractor fails to install and maintain a satisfactory standard of Site facilities, Siemens Energy reserves the right to install and maintain such facilities at Contractor's expense.
- 44.7. Where the Contractor must install hygiene and sanitary facilities such as; toilets, portable/chemical toilets, hand basins, showers/washrooms and locker rooms, these shall be:
 - 44.7.1. Maintain separated male and female

- 44.7.2. Be fully adequate in size, number and configuration following local regulations.
- 44.7.3. With an adequate supply of hot/cold water to service the number of personnel;
- 44.7.4. Showers and toilets partitioned from each other and have lockable doors for privacy;
- 44.7.5. Facilities must be ventilated and provided with adequate lighting and heating where needed;
- 44.7.6. Do not open directly into workrooms or mess rooms.
- 44.7.7. Kept clean and promote hygienic conditions.

45.

Steel Erection

- 45.1. A site-specific Steel Erection Plan shall be provided to Siemens Energy before steel erection activities begin. This plan shall be prepared by a Qualified person and shall address the following, at a minimum:
 - 45.1.1. Working at elevation procedures for the erection process
 - 45.1.2. Training of workers involved with the steel erection process
 - 45.1.3. Erection sequence
 - 45.1.4. Crane selection and placement
 - 45.1.5. Overhead protection / routing of lifts
 - 45.1.6. Procedures for steel erection activities (e.g., bracing / guying, connections, decking, roofing, siding, grating, etc.)
 - 45.1.7. Perimeter fall protection planning and turnover
- 45.2. All columns shall be anchored by a minimum of four anchor bolts.
- 45.3. Anchor bolts shall not be repaired, replaced, or field modified without the approval of Siemens Energy.
- 45.4. All steel members shall be secured by a minimum of two bolts per connection prior to un-hooking steel member from crane.
- 45.5. Purlins and girts shall not be used as anchorage points for fall arrest systems unless written approval is obtained from a Qualified person.
- 45.6. Walking on I-beams shall not be permitted during steel erection activities unless acceptable measures (e.g., engineered horizontal lifelines or safety nets) are implemented as agreed upon by Siemens Energy.
- 45.7. The use of beamers, cheaters, or other fall protection devices that permit workers to freefall greater than 6 feet in the event of a fall shall not be considered an acceptable control measure.

46.

Storm and Lightning

- 46.1. Contractor shall develop a Severe Weather Plan and submit to Siemens Energy for review and acceptance and shall consider at a minimum:

- 46.1.1. Weather monitoring, notification to affected personnel, safe shelters for workers, and suspension / resumption of work affected by such weather conditions.
- 46.1.2. Identify and explain the systems or devices used to track and monitor approaching storms, effective distances for alerting personnel and suspending work, duties of persons responsible for plan implementation, means of communication for alerting site personnel, and stay times and/or distances before resuming work.
- 46.1.3. Effective distances for work suspension, especially for elevated work and crane activities that provides personnel adequate time to stop work, secure items, and be inside a safe shelter.
- 46.1.4. At a minimum, effective distances and stay times shall meet or align with those published by a recognized source or creditable organization (e.g., NOAA, National Lightning Safety Institute, OSHA, ANSI, CDC / NIOSH, federal / state agency, or etc.).

For example, all personnel shall be inside a safe shelter before the storm is within 10 miles and work may not resume until 30 minutes after the thunderstorm passes (i.e., last sign of lightning).

47.

Vehicles and Equipment

- 47.1. Contractor shall develop and maintain a project specific "Site Traffic Management Plan" that describes road signage and delineation, parking arrangements, speed limits, and mobile equipment / vehicular traffic segregation from pedestrians.
- 47.2. A site map shall be designed and integrated with the Project's Traffic Management Plan. Site map shall be updated and kept current as the Project evolves.
- 47.3. Site map should identify key locations, such as offices, medical station, work areas, parking areas, site roads, pedestrian crossings, direction of travel, fuel storage or chemical tanks, truck staging areas, laydown yards, evacuation routes, and rally points/ muster locations.
- 47.4. Site maps shall be provided or posted in key locations (e.g., guard shack, site entrance, and/or site bulletin board) for reference by workers, visitors, and security personnel.
- 47.5. Modifications to the Site Traffic Management Plan shall be requested in writing by Contractor and provided to Siemens Energy for review and acceptance.
- 47.6. Contractor shall control mobile equipment movement on the Project Site by use of flaggers and/or spotters, traffic control devices, appropriate signage, and worker training.
- 47.7. Contractor shall reverse park all vehicles on the project premises, unless otherwise accepted by Siemens Energy.
- 47.8. Mobile equipment and vehicles traveling in reverse, in congested areas, or near plant equipment shall have an audible back-up alarm and Spotter at all times.
- 47.9. Spotters and flaggers shall be appropriately trained and Qualified to perform the work, and they shall use a standard set of hand signals. Training documentation shall be available and provided on site.
- 47.10. Spotters shall maintain an adequate distance (i.e., separation) from mobile equipment.

Subject to Management Review and Approval

At a minimum, the following distances shall be observed:

- 47.10.1. At least of 25 feet shall be maintained when in front or behind construction vehicles and mobile equipment (e.g., forklifts, telescoping handlers, pickup trucks, light-duty vehicles, and skid steer loaders), which are smaller in size.
- 47.10.2. At least two (2) vehicle lengths shall be maintained from earth moving equipment (e.g., scrapers, graders, loaders, bulldozers, drum compactors, dump trucks, etc.), which is larger in size with bigger blind spots.
- 47.11. Spotters shall wear appropriate high visibility garments in accordance with ANSI/ ISEA 107 and other necessary PPE per the hazard assessment for the task.
- 47.12. Visual contact (i.e., line of sight) shall be maintained at all times between the Spotter and mobile equipment operator. Equipment movement shall stop immediately if visual contact is lost. Visual contact with the Spotter shall be re-established before mobile equipment movement may resume.
- 47.13. No personnel shall be transported in the back of vehicles (e.g., pickup truck beds). Passengers are only permitted to ride in vehicles with a designed passenger cab, seat(s), and seatbelt(s).
- 47.14. Construction vehicles and equipment shall be equipped with a back-up alarm that sounds when the vehicle is reversing. Back-up alarms shall be fully functional and audible above the surrounding noise level.
- 47.15. Training for earthmoving equipment, aerial lift, skid steer loader, and powered industrial truck operators shall be performed in accordance with OSHA 1926 and 1910 standards.
 - 47.15.1. Equipment operator training shall include documented classroom training (i.e., formal instruction) and field practical examination (i.e., operator evaluation).
 - 47.15.2. Contractor shall provide and maintain an up-to-date register of all truck drivers and Qualified mobile equipment operators that enter the site.
- 47.16. Contractor shall submit a formal request to Siemens Energy prior to placing mobile equipment (e.g., skid steers, scissor lifts, articulating man lifts, etc.) on permanent plant elevated structures (e.g., mezzanine floors, turbine decks, and grated elevations).
- 47.17. All drivers / operators shall wear seatbelts, and they shall ensure their passengers do the same. All passengers shall be secured (e.g., doors closed and seat belts fastened) before moving the vehicle.
- 47.18. Drivers shall follow the speed limits, traffic signs, and maintain a safe following distance at all times (i.e., no tailgating).
- 47.19. Texting, reading texts, or using of other electronic devices (e.g., laptop computers, tablets, and palm handheld devices) is prohibited while driving or operating vehicles or mobile equipment.
- 47.20. Talking on a hand-held cell phone (even with a hands-free device) while driving a commercial motor vehicle or operating mobile equipment is prohibited. The use of two-way radios is permitted when operating mobile equipment or driving a vehicle.
- 47.21. The use of personal entertainment devices, including headphones, are prohibited while operating equipment.
- 47.22. The use of golf carts is prohibited on the project site unless otherwise accepted by Siemens Energy.
- 47.23. Side-by-Side/ Utility Task Vehicles (UTVs) shall not exceed 10mph. Contractor shall

implement control measures to prevent speeding.

47.24. At a minimum, the following the safety equipment / devices shall be installed on all onsite vehicles including all side-by-side / utility task vehicles (UTV's) and pick-up trucks (purchased or rented) used on the project:

47.24.1. Seatbelts

47.24.2. Headlights (i.e., required for nighttime and/or low light conditions)

47.24.3. Roll-Over Protection System (ROPS) (except pick-up trucks)

47.24.4. Back-Up Alarm

47.24.5. Fire Extinguisher

47.24.6. Flashing / Strobe Lights (Roof-mounted flashing / strobe lights and/or "buggy whips / flags" are recommended on UTV's for improved detection / visibility, especially by drivers of larger vehicles.)

47.25. Contractor shall survey all power lines and overhead structures (e.g., bridges, conveyors, and pipe racks or bridges) on site.

48.

Work Zone Safety / Public

Roads

48.1. When working on, near, or adjacent to public roads, physical barriers (e.g., hard barricades / concrete "Jersey" barriers and/or parked / stationary equipment and vehicles) shall be utilized and positioned to shield workers and create a buffer zone between the workers and the direction of vehicular traffic.

48.2. Additional soft barricades (e.g., traffic cones, traffic barrels, or A-frame horses) shall be used in addition to the hard barricades to delineate and define the Safe Zone around workers.

48.3. Any additional controls required by the U.S. Department of Transportation (DOT)/ Federal Highway Administration's Manual on Uniform Traffic Control Devices (MUTCD) and/or local agencies shall also be implemented when working near public roads (i.e., in or near the road right-of-way).

48.3.1. For short-term operations that are not covered by federal, state, county, city, or local traffic regulations, workers may use rotating, flashing, or strobe lights on vehicles instead of signs. However, workers shall place reflective traffic cones along the full length of the vehicle and taper the cones toward oncoming traffic.

48.4. Flaggers shall be properly trained and wear ANSI Class 3 apparel while performing flagging duties and use reflective sign paddles or flags according to the jurisdictional and Department of Transportation requirements.

OCCUPATIONAL HEALTH

49. Asbestos Containing Materials

- 49.1. Asbestos Containing Materials (ACM) are prohibited on all Siemens Energy projects.
- 49.2. If there is a potential for contacting or disturbing existing ACM, Contractor shall develop plans to control exposure and protect employees in accordance with Federal, state and local regulations.

50. Avoiding Insects and Snakes

- 50.1. Contractor shall assess the potential for encounters with snakes, insects, wasps, yellow jackets, ants, and ticks.
- 50.2. Based on the assessment, Contractor shall have a mitigation plan to control hazards and protect employees.

51. Blood-borne Pathogens

- 51.1. Contractor shall train each employee trained to provide first aid, on the recognition and avoidance of unsafe conditions and how to control or eliminate any hazard or other exposure to illness or injury from blood-borne pathogens.
- 51.2. Contractor shall develop and implement a Blood-borne Pathogens (BBP) Program which is designed to prevent contact with the eyes, mouth, mucous membranes and non- intact skin with blood or other potentially infectious body fluids, when a first aid room is part of their site facilities and installations.

52. Electromagnetic Fields (EMF) and Medical Devices

- 52.1. High electromagnetic fields (EMF) may interfere with the operation of medical devices. High EMF levels may be present in electrical equipment located in generating station switchyards, station main bus lines, magnetic separators, generator brush rooms and other locations such as substations. Office areas do not present a risk.
- 52.2. Workers who have medical devices such as internal pacemakers, defibrillators, insulin pumps and pain modification devices are encouraged to consider voluntarily reporting such devices to supervision.
 - 52.2.1. Management cannot require workers to report medical devices, but they should consider doing so.
 - 52.2.2. While cochlear implants and hearing aids may be affected, they should not

present a life-threatening risk.

- 52.3. Personnel with medical devices should not enter generating plant switchyards due to potential for high electromagnetic fields.

- 52.4. If there appears to be an immediate hazard related to medical device operation and high levels of EMF:
 - 52.4.1. Stop work.
 - 52.4.2. Move away from the high EMF source.
 - 52.4.3. Contact your supervisor.

53.

Eyewash and Emergency

Showers

- 53.1. Contractor shall provide eyewash and emergency showers as needed based on site conditions, work environment, and potential hazards that could be encountered.
- 53.2. Eyewash stations and emergency showers shall be located as close to the potential exposure location as possible.
- 53.3. Workers shall be trained in the proper use of eyewash stations and safety showers.

54.

Hazard Communication

- 54.1. Contractor shall notify Siemens Energy of all chemicals and hazardous materials at least fourteen (14) days in advance of bringing them onto the Project Site. Contractor may be prohibited from bringing some chemicals on Project Site if the chemical or material has been disapproved by the site EHS Professional or banned by Siemens Energy.

Contractor shall provide Siemens Energy a current Safety Data Sheet (SDS), estimated amount or quantity of material brought on Project Site, proposed days on the Project Site, storage location, and any special handling / storage requirements for each chemical or hazardous material.
- 54.2. Contractor shall prepare an index listing of chemicals to be used on the Project Site (i.e., Hazardous Chemical List) shall be created, maintained, and updated when a new Safety Data Sheet (SDS) is added by the contractor.
- 54.3. Contractor shall maintain SDSs on Project Site for all chemicals brought on by them and their Subcontractors. Further, the Contractor shall provide SDS access available to all of their employees, Subcontractors, and Siemens Energy personnel.
- 54.4. Contractor shall use only approved chemical containers, and all containers shall be labeled.
- 54.5. Contractor is responsible for and must remove all chemical commodities from the Project Site that are not consumed during the job.

55.

Hearing Conservation

- 55.1. Hearing protection shall be worn while working in all areas where the noise level is equal to or greater than 85 dB and in areas where workers cannot carry on a conversation in normal tones within 2 to 3 feet of each other.
- 55.2. Hearing protection shall be worn according to the instructions on signs and labels

Subject to Management Review and Approval

identifying potential high-noise exposure.

- 55.3. Contractor shall implement a hearing conservation program, in accordance with applicable regulations, to ensure workers are protected from hearing loss. In all cases where the sound levels exceed the values shown below, a continuing, effective hearing conservation program shall be administered.

PERMISSIBLE NOISE EXPOSURES	
Duration Per Day, Hours	Sound Level Dba Slow Response
8	90
6	92
4	95
3	97
2	100
1.5	102
1	105
.5	110
.25 or less	115

56.

Heat and Cold Exposure

- 56.1. Contractor shall implement a heat illness prevention program to ensure workers are not adversely affected by heat stress.
- 56.2. Contractor shall implement a program to mitigate the effects of cold temperature and wind exposures.
- 56.3. Contractor shall provide each program to Siemens Energy at least 30 days prior to exposure to hot or cold temperatures.
- 56.4. Snow, ice, mud and water shall be removed from equipment and working surfaces when a hazard is present.
- 56.5. UV protection for eyes and skin should be used when necessary.
- 56.6. Contractor shall include the following elements in the program
- 56.6.1. The systems and devices used to track and monitor heat index and wind chill at the project (e.g., ambient temperatures and humidity), temperatures or criteria (i.e., benchmarks) for implementing controls (e.g., work / rest schedules and hydration targets), duties of persons responsible for plan implementation, and means of communication for alerting site personnel.
 - 56.6.2. Effectively plan, engineer and schedule work/projects to eliminate, control or minimize exposure to temperature extremes and include, describe, and address preventative measures and specific controls used to minimize the potential for heat / cold related illnesses (e.g. heat exhaustion, heat stroke, frostbite, hypothermia, etc.).
- 56.7. Take adequate breaks and replace fluids frequently.
- 56.7.1. At a minimum, work / rest schedules and control measures shall meet or align with those published by a recognized source or creditable organization (e.g., OSHA, ANSI, CDC / NIOSH, ACGIH, federal / state agency, or etc.).

- 56.8. Maintain good physical condition to minimize and/or eliminate injuries/ illnesses from exposure to heat or cold.
- 56.9. When working in adverse weather conditions (e.g., snow/ice storms), take appropriate precautions.
- 56.10. Wherever possible, utilize the buddy system if you must travel or work in extreme temperatures.
- 56.11. When working (or preparing to work) in areas of extreme heat, consume adequate and appropriate amounts of fluids to help avoid heat-related illnesses and dehydration.
- 56.12. Adequately replace salt and other minerals through normal diet. (Supplements are usually unnecessary.)
- 56.13. Plan work based on recommended stay times. Provide adequate personnel for rotation.
- 56.14. Slowly build up a tolerance for heat exposure. Where appropriate, reduce stay time accordingly.
- 56.15. When working in hot environments, wear appropriate clothing. Use ice vests if desired.
- 56.16. Take precautions to protect from overexposure to the sun (e.g., clothing, eye protection, sunscreen).
- 56.17. Permit workers to interrupt their work if experiencing signs of heat- related disorders or extreme discomfort.
- 56.18. Observe co-workers for symptoms of heat-related illness. When necessary, seek medical attention. Symptoms include but are not limited to dry, hot skin and high body temperature (heat stroke), moist clammy skin (heat exhaustion) and heat cramps.
- 56.19. When working in cold environments, dress to stay warm and dry. Use rain gear, layered or insulated clothing as appropriate.
- 56.20. Observe co-workers for symptoms of frostbite and hypothermia. When necessary, use warming shelters or vehicle cabs for temporary relief. Seek medical attention when appropriate.
- 56.21. Walk carefully on snowy and icy surfaces.
- 56.22. When shoveling snow, be very careful to avoid overexertion.
- 56.23. Soaking wet clothing contributes to hypothermia and should be removed as soon as possible.
- 56.24. Use of vibrating tools in extremely cold weather could aggravate a circulatory condition called Raynaud's Syndrome and should be avoided by affected individuals.
- 56.25. Avoid heat-related illness by maintaining adequate levels of hydration. This table provides general length of time between water breaks and corresponding hydration target for each hour of work.

Temperature	WorkLevel	Maximum Minutes Worked Between HydrationBreaks	Hydration Target
<80	Normal		8-12 oz./hr.
80-85	Normal		8-16 oz./hr.
86-90	Normal	50	12-20 oz./hr.
91-95	Normal	45	16-24 oz./hr.
>96	Normal	40	24-32 oz./hr.

57. Hexavalent Chromium

- 57.1. Contractor shall identify all activities with the potential to produce exposure to hexavalent chromium.
- 57.2. If the potential for hexavalent chromium exposure exists, Contractor shall develop a plan to control exposure and protect employees in accordance with federal, state and local regulations.

58. Lead in Construction

- 58.1. Lead-based paints and coatings are prohibited on Siemens Energy projects.
- 58.2. Contractor shall identify all activities with the potential to produce exposure to lead.
- 58.3. If there is a potential exposure for lead, Contractor shall develop a plan to control exposure and protect employees in accordance with federal, state and local regulations.

59. Radiography

- 59.1. The companies carrying out the tests and examinations shall take all actions stipulated in the license for handling radioactive materials and X-ray tubes and shall guarantee this in writing to the Site Management.
- 59.2. Before any radiography is performed at the site, a meeting is required between Site Management and the radiographic contractor. In this meeting the location, time, and radiographic test to be performed will be established in writing.
- 59.3. Radiographies will be previously communicated to all personnel with drawing indicating time, affected areas and safe boundaries and will be posted in a conspicuous place.
- 59.4. A suitable dosimeter equipment is available on site.
- 59.5. Contractor shall supply appropriate personnel monitoring equipment, such as film badges, pocket chambers, pocket dosimeters, or film rings to every employee who enters a restricted area.
- 59.6. Radiation restricted areas must be established by the radiographer before exposing the radiation source.
- 59.7. Restricted areas must be identified by magenta and yellow "DANGER RADIATION" signs and tape. Radiation signs should be posted in English and Spanish if the workforce includes non-English speaking persons. The barricades and signs shall meet

US NRC requirements. As a minimum, barricades and signs should be posted at a 2 MREM estimated exposure level for radiographic examinations

- 59.8. Only authorized individuals are permitted inside established radiation restricted areas.
- 59.9. Authorized individuals entering radiation restricted areas must be appropriately trained and have personnel radiation dosimeters (e.g., film badges, alarming rate meters).
- 59.10. During any calendar quarter the dose to the whole body shall not exceed 3 rems; and the dose to the whole body, when added to the accumulated occupational dose to the whole body, shall not exceed 5 (N-18) rems, where "N" equals the individual's age in years at his last birthday
- 59.11. Site Management will communicate radiographic testing specifics to site personnel.
- 59.12. Operating and supervisory contractor personnel are in attendance at all times during the tests and examinations
- 59.13. Radioactive isotopes are properly placed and secured inside the shielding containers after completion of the tests and examinations in accordance with US NRC requirements.
- 59.14. The X-ray unit is switched off properly after completion of the tests and examinations.

60. Respiratory Protection

- 60.1. Contractors Environmental, Health, and Safety Plan shall address the following:

Respiratory hazards, which include hazardous dusts, mists, vapors, gases, fumes, smoke, biological hazards (e.g., bird feces) or any hazardous particulate. Except where voluntarily used for nuisance dust, the selection and use of respirators shall be based on actual air sampling data or a calculated estimate of exposure levels above the regulatory exposure limits in 29 CFR Subpart Z. Respiratory protection level and cartridge type (where used) shall be appropriate for the chemical hazard encountered and for the anticipated or known exposure level.

Operations that generate dust from materials containing silica (e.g., sawing, drilling, or cutting concrete, refractory work, or vacuuming fly ash or coal) shall be presumed to exceed OSHA's Permissible Exposure Limit (PEL) for crystalline silica without the use of adequate controls. Protective measures, such as wet methods and HEPA vacuuming, shall be used to reduce worker's exposure levels below the PEL.

- A written exposure control plan shall be submitted to Siemens Energy for acceptance describing control methods to be developed and implemented for tasks addressed in 29 CFR 1926.1153, Table 1.
- Control methods (e.g., engineering controls, work practices, and respiratory protection) for activities addressed in 29 CFR 1926.1153, Table 1 shall be followed.
 - If controls methods identified in 29 CFR 1926.1153, Table 1 are not implemented, Contractors shall demonstrate, at no cost to Siemens Energy, that worker exposure levels are below the PEL in accordance with in 29 CFR 1926.1153 (d).

- If exposure monitoring data demonstrates there is an overexposure to personnel, additional controls methods shall be implemented, at no cost to the Siemens Energy, to reduce worker exposure levels below the PEL.

A complete environmental, health, and safety plan is required when working with or generating hexavalent chromium (e.g., welding or torch cutting stainless steel), inorganic arsenic (e.g., welding or torch cutting boiler components with significant scaling or vacuuming of fly ash), lead, silica dust, man-made vitreous fibers, and/or asbestos.

- 60.2. Contractor is responsible for anticipating, recognizing, evaluating, and controlling factors in the workplace that may cause sickness, impaired health / wellbeing, or significant discomfort among workers. Contractor shall utilize Qualified persons, such as an Industrial Hygienist, for guidance in addressing workplace hazards capable of causing illnesses, such as, but not limited to the following:

Exposures to hexavalent chromium, lead, silica, arsenic (e.g., fly ash at coal plant sites), vanadium (e.g., ash at oil fired sites), asbestos, man-made vitreous fibers (e.g., refractory ceramic fibers – RCF), thermal stress, high noise, ergonomic hazards, and etc.

Where industrial hygiene (IH) monitoring is performed by Contractor or third party, a copy of the IH monitoring results shall be provided to Siemens Energy. Names may be removed to preserve privacy concerns. Industrial hygiene air samples shall be submitted to a laboratory accredited by the American Industrial Hygiene Association for the analytical method used. Industrial hygiene monitoring shall be performed by a third party not directly employed by Contractor, unless otherwise accepted by Siemens Energy.

- 60.3. Siemens Energy reserves the right to perform industrial hygiene monitoring on site personnel to verify worker exposure levels and/or help ensure adequate controls are used. Monitoring results will be provided to Contractor and/or appropriate Subcontractor. However, the duty to perform IH surveys to ensure regulatory compliance (i.e., burden of proof) resides with the respective Contractor performing the work (e.g., workers' employer) and shall be performed at no cost to Siemens Energy.

- 60.4 Engineering controls (e.g., ventilation) and/or PPE shall be used to adequately control worker exposure levels below recognized / acceptable limits. Unless permitted by an OSHA standard, Contractor shall not use job rotation as an administrative control to keep workers' exposure levels to airborne contaminants below a permissible limit (e.g., asbestos, hexavalent chromium).

61.

Substance

Abuse

Prevention

- 61.1. Contractor shall implement and provide to Siemens Energy a Substance Abuse Program designed to promote a drug and alcohol-free work environment. The Contractor shall, as part of their ongoing drug and alcohol abuse prevention program, perform testing on all persons assigned to work on the project in the following instances.

Pre-employment

61.1.1.1. Each employee's employer is responsible for conducting pre-employment testing. A formal letter listing the names of personnel who have successfully completed the pre-employment testing must be submitted to Siemens Energy.

Random
Reasonable Suspicion
Post Incident

61.1.1.2. Post incident screening is to be performed in the event of an injury requiring treatment beyond basic first aid, property damage, vehicle accident, near miss event, or at the request of Siemens Energy.

61.2. Any person found in possession of illegal substances (drugs), drug paraphernalia, or alcohol shall be permanently removed from the project site.

61.3. Alcohol Testing

Any person with a blood alcohol level of 0.02 or higher shall be immediately removed from the site for a period of 24 hours and retested before beginning work again. Any second offense will result in permanent removal from the site.

Saliva Alcohol Test is acceptable for initial alcohol testing; however, confirmation must be performed using methods which provide legally admissible results.

61.4. Drug Testing

The levels of acceptance (rejection) for drugs shall be determined based upon NIDA guidelines for a NIDA 5 panel screen using current NIDA cutoff and confirmation levels for the following 5 drug groups.

Amphetamines
Cocaine
Cannabinoids
Opiates
Phencyclidine

61.5. Any person with a confirmed non-negative drug screen result shall be permanently prohibited access to the site. Non-negative results shall be confirmed via gas chromatography/mass spectrometry (GC/MS) laboratory analysis and the use of a Medical Review Officer (MRO).

61.6. Drug testing requirements may vary due to specific laws in the country where the project is located. This section may be modified, based on specific laws.

62.

Welding and Thermal

Cutting – Health Exposures

62.1. Do not clean with compressed air. Use HEPA vacuums or other means that do not disperse dust into the air.

62.2. Do not take contaminated protective clothing or equipment home.

62.3. Do not enter eating/drinking area with work clothing unless dust has been removed (in a manner that does not disperse dust into the air).

62.4. Whether working in the weld shop or in the field, position yourself so that your breathing zone is not in the fume plume, regardless of the relative hazard of the process. If work conditions prevent proper positioning, use ventilation to direct fumes away from welders, cutters and helpers.

62.5. When using welding screens around your work, arrange screens so that restrictions to ventilation are minimized.

- 62.6. Position the exhaust system take-off near enough to the electrode to assure maximum fume capture when using local exhaust ventilation.
- 62.7. Do not use oxygen for ventilation purposes.
- 62.8. Where ventilation or natural conditions do not control welding/cutting fumes from reaching the workers' breathing zones, use appropriate respirators.
- 62.9. When possible, remove all coatings by chemical or mechanical means (with local exhaust ventilation) prior to thermal cutting. Remove a "cut line" of at least a width of 3 to 4 inches. If coating removal is not possible, use ventilation and/or respiratory protection.
- 62.10. Prior to removing or disposing of coating materials, determine if the coating contains lead or hexavalent chromium. See your location H&S/ EHS professional for guidance.
- 62.11. Before applying heat, verify that metal is clean of any solvents (particularly chlorinated solvents), oils or other materials that may cause hazardous decomposition products.
- 62.12. Do not perform welding or thermal cutting activities within 200 feet of the storage of chlorinated solvents.

Discuss potential fume hazards in the pre-job brief. Review product SDS and/or hazard label, ventilation options and PPE. Coordinate with H&S/ EHS professional to determine if adequate exposure assessments have been completed for the specific activity.

ENVIRONMENTAL PROTECTION

63. Environmental Plans

- 63.1. Contractor shall inspect the construction of erosion and sediment control measures and submit written certification to Siemens Energy, bearing his or her Engineer of Record (EOR) professional seal, stating the measures have been built in conformance with the approved plans and specifications prior to commencement of any land disturbing activity. Outside USA the equivalent engineer to EOR will be accepted.
- 63.2. Contractor shall be responsible for performing and documenting erosion and sedimentation-related (i.e., storm water) inspections, at least weekly and after each significant rain event, as per permit requirements. Reports shall be kept and made available to Siemens Energy for review and acceptance within 24 hours of the inspection covered by the report and prior to the report being finalized. Siemens Energy shall retain copies of finalized forms for record.
- 63.3. Temporary fuel and chemical storage tanks shall be double walled or have outside containment that will contain the full volume of the tank plus 10% additional overboard. Containment shall be a lined berm or manufactured pop-up containment.

64. Spill Prevention

- 64.1. Protection shall be provided for drips, leaks, and spills at all temporary fuel and

chemical storage areas or tanks, including double walled tanks. The Contractor is responsible for registering the tank, if required by law, and maintaining any required Spill Prevention, Control, and Countermeasure (SPCC) Plans unless otherwise agreed to by Siemens Energy.

- 64.2. EPC / Prime / General Contractor shall create and provide Siemens Energy a site map, aerial photograph with markings, or drawing from the Erosion & Sediment Control Plan. The document shall identify and clearly show the agency-approved Limits of Disturbance (LOD), all wetlands on and encompassing the Project site, and any surface waters.
- 64.3. Contractor shall submit an Equipment Storage and Spill Mediation / Recovery Plan to Siemens Energy for review and approval. The plan shall address mobile, portable, and stationary equipment with diesel fuel, gasoline, hydraulic fluid, oil, chemicals, or other petroleum products. This is a separate plan in addition to any Spill Prevention Control and Countermeasures (SPCC) Plan that may be required.
- 64.4. Mobile equipment shall have drip protection (e.g., drip pan, absorbent pad / diaper, poly sheeting, etc.) provided beneath the equipment at all times when the equipment is parked.
- 64.5. Mobile equipment shall have spill kits, absorbent pads / diapers, or similar materials on or inside the equipment at all times.
- 64.6. Contractor shall train personnel assigned to the Project on spill prevention requirements, location and use of spill kits and containment measures, reporting expectations for leaks, spills, and chemical releases, and maintenance of additional environmental protection measures (e.g., silt fencing, berms, catch basins, and storm drain protection).
- 64.7. Portable equipment (e.g., lights plants, generators, welding machines, and air compressors) shall have drip protection and containment provided at all times beneath the equipment.
- 64.8. Stored equipment, tanks, and containers with hazardous chemicals or petroleum products shall have a spill containment system in place to prevent any spilled or leaked fluid from reaching storm drainage, runoff, or water bodies.
 - 64.8.1. All containments shall be evaluated for evidence of product (i.e., sheen) prior to being drained or emptied of rainwater. Contractor shall notify the Siemens Energy site environmental specialist, EHS professional, or designated person prior to emptying or draining containments.
- 64.9. Hauls trucks, material / equipment delivery trucks, mobile equipment, and construction vehicles (e.g., earth moving equipment, drill rigs, material handling equipment, etc.), and construction equipment (e.g., light plants, portable generators, air compressors, and aerial lifts) shall be visually inspected for leaks (e.g., fuel, oil, hydraulic fluid, coolant, etc.) prior to initially entering the site. Equipment that is leaking fluids, unsafe, or not properly maintained is prohibited from entering the Siemens Energy site.
 - 64.9.1. Construction equipment and vehicles must be maintained and leak-free while on site. Equipment or vehicles that experience multiple problems, deficiencies, or occurrences shall be removed from site and replaced at the Contractor's expense.
- 64.10. Vehicles and mobile equipment shall be parked greater than 125 feet from surface waters and storm drains. When this is not possible, Contractor shall ensure

preventative measures are in place to prevent all released petroleum products and other fluids from reaching the ground, storm drains, and/or "Waters of the United States."

- 64.11. Refueling shall be conducted greater than 125 feet from surface water and storm drains and requires 100% attendance by the fueling attendant. The fueling device/ truck shall have appropriately rated fire extinguishers in adequate quantity for the fuel being dispensed and a suitable spill kit to immediately remediate any release.
- 64.12. All fueling stations / trucks shall have appropriate signage and labeling required by OSHA, NFPA, DOT, and/or other government agencies as well as "No Smoking Within 25 Feet" signs shall be posted in all locations where flammable liquids are dispensed, used, or stored.
- 64.13. For permit modifications initiated by the permittee, the Contractor shall give notice to Siemens Energy, as soon as possible, regarding any planned modifications, alterations, or additions to environmental permits so the Siemens Energy Environmental Services department can be informed.
- 64.14. Any spill or release related to the Project Site, which causes a film or sheen upon or discoloration of the surface of the water or adjoining shorelines or causes a sludge or emulsion to be deposited beneath the surface of the water or upon adjoining shorelines, is reportable to the governing regulatory agency.
 - 64.14.1. EPA guidance indicates that a spill reported within 15 minutes is deemed to be reported "immediately." Therefore, every effort should be made to report within 15 minutes any fuel / oil or chemical spill to surface water, soil within 100 feet of water (i.e., "Waters of the United States"), or releases >25 gallons to the Siemens Energy onsite EHS Professional or facility designee.
 - 64.14.2. Notification of spills or releases, which are reportable to a regulatory agency, shall be made by Siemens Energy. In other words, Siemens Energy shall contact the regulatory agency and provide notification to the appropriate governing bodies of the reportable spill or release. Therefore, it is essential all spills and releases, regardless of size and location, be immediately reported to Siemens Energy.
 - 64.14.3. Contractor shall provide to Siemens Energy a recovery plan with corrective actions following an environmental incident that requires notification to a regulatory agency.
- 64.15. Contractor shall immediately notify Siemens Energy of any exceedance of stated limits or parameters specifically identified in environmental permits or violation of an environmental law or regulation discovered as the result of an internal audit, self-assessment, or observation.

65.

Clean up and Disposal

- 65.1. Contractor is responsible for the clean-up and proper handling, collection, containerizing, and removal of regulated wastes, as defined by the EPA and state or local regulations.
 - 65.1.1. Contractor shall handle, store, and manage all regulated wastes in accordance with federal, state, and local laws and regulations and Siemens Energy requirements, which may vary depending on the state, county, local municipality, and Project Site.

- 65.1.2. Regulated waste, including to but not limited to, hazardous waste, used oil, oily debris, universal waste, and non-hazardous waste shall be disposed by the responsible party as stated in the terms and conditions of the Agreement.
- 65.1.3. Contractor shall consult with Siemens Energy for approved vendors and landfills for the disposal of regulated wastes, including to but not limited to, used oil, oily debris, universal waste, and non-hazardous waste.
- 65.1.4. Contractor shall provide copies of signed manifests for waste shipments and waste profiles and maintain associated paperwork.
- 65.2. Upon or near Project completion, a formal review of all environmental-related permits/ plans shall be completed with the Contractor and the Siemens Energy to review the status of all environmental permits, plans, inspection requirements, and/or agency notifications.

ADMINISTRATIVE DOCUMENTS

66. Cross References

None

67. Standards

OSHA 1926	Construction Standards
ANSI / ASME B30.20	Crane and Rigging
ANSI / ISEA	High Visibility Clothing
ASME P30	Planning for Load and Handling Activities
ANSI A14.5-2000	Portable Reinforced Plastic Ladders (American Ladder Institute)
ASTM F1790	Standard Test for Measuring Cut Resistance
NFPA 30	Standard for Flammable Liquids Safety
NFPA 101	Life Safety Code
NFPA 70	National Electrical Code (NEC)
NFPA 70E®	Standard for Electrical Safety in the Workplace®
NFPA 10	Standard for Portable Fire Extinguishers
EM 385-1-1	U.S. Army Corps of Engineers' Safety and Health Requirements Manual

Subject to Management Review and Approval